

TOPIC NAME : _____

TIME : _____

DATE : / /

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Final Term
Review

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Partial Derivatives

function of Several Variables

laws- if, $z = f(x, y)$, then

1. $z_x = f_x = \frac{\partial f}{\partial x} \rightarrow$ Partial differentiation of with respect to x , taking y as constant

2. $z_y = f_y = \frac{\partial f}{\partial y} \rightarrow$ Partial differentiation of with respect to y , taking x as constant

types 1:-

Ex:- $f(x, y) = 2x^2y^2 + xy^2 - 3x - 4y + 4$

$$z_x = f_x = \frac{\partial f}{\partial x} = 4xy^2 + y^2 - 3 + 0 + 0 =$$

$$z_y = f_y = \frac{\partial f}{\partial y} = 4x^2y + 2xy - 4$$

Law

3. $f_{xx} = (f_x)_x$

4. $f_{yy} = (f_y)_y$

5. $f_{xy} = (f_x)_y$

6. $f_{yx} = (f_y)_x$

7. $f_{xxy} = (f_{xx})_y$

8. $f_{yyx} = (f_{yy})_x$

Q:- $f_{xx} = ?$

$\Rightarrow f_x = 4xy^2 + y^2 - 3$

$\Rightarrow$ so, $(f_x)_x = 4y^2$

Q:- $f_{xy} = ?$

$\Rightarrow f_y = 4x^2 + 2xy$

$\Rightarrow$ so, $(f_y)_x = 4x + 2y$

Q:- $f_{xx} = ?$

$\Rightarrow (f_x)_x = 4y^2$

$\Rightarrow$ so, $(f_{xx})_x = 0$

Q:- $f_{xy} = ?$

$\Rightarrow (f_y)_x = 4x + 2y$

$\Rightarrow$ so, $(f_{xy})_y = 0$

Q:- $f_{xy} = ?$

$\Rightarrow f_x = 4xy^2 + y^2 - 3$

$\Rightarrow$ so, $(f_x)_y = 8xy + 2y$

Q:- $(f_y)_x = ?$

$\Rightarrow f_y = 4x^2 + 2xy$

$\Rightarrow (f_y)_x = 8xy + 2y$

Chebyshev's Theorem -

Law 1 -

1. $f_{xy} = f_{yx}$
2. $l_{xy} = l_{yx}$

Q:- $f_{xyx} = (?)$

$$\Rightarrow (f_{xy})_x = 8y + 0$$

$$= 8y$$

Q:- $f_{yxy} = (?)$

$$\Rightarrow (f_{yx})_y = 8x + 2$$

Q:- $f_{yxx} = (?)$

$$\Rightarrow (f_{yx})_x = 8y + 0$$

$$= 0 + 8y$$

$$= 8y$$

$$f(x,y) = x^2 + y^2$$

$$f_{xx} = 2x$$

$$f_{xy} = 2y$$

$$f_{yy} = 2y$$

$$f_{xyx} = 2$$

$$f_{yxy} = 2$$

$$f_{yxx} = 2$$

$$(2)(1)(1) = 2$$

$$f_{xyx} = 2$$

$$f_{yxy} = 2$$

$$f_{yxx} = 2$$

$$f_{xyx} = 2$$

$$f_{yxy} = 2$$

$$f_{yxx} = 2$$

Pages- 0248-

Find the 1st partial derivatives of the functions

(15) $f(x, y) = x^4 + 5xy^3$

$$\Rightarrow f_x = 4x^3 + 5y^3$$

$$f_y = 15xy^2$$

(16) $f(x, t) = t^2 e^{-x}$

$$\Rightarrow f_x = -t^2 e^{-x}$$

$$\Rightarrow f_t = 2te^{-x}$$

(17) $z = \ln(x + t^2)$

$$\Rightarrow z_x = \frac{1}{x + t^2} \times 1$$

$$= (x + t^2)^{-1}$$

$$z_t = \frac{1}{x + t^2} \times 2t$$

$$= \frac{2t}{x + t^2}$$

$$= 2t(x + t^2)^{-1}$$

$$\textcircled{40} u(r, \theta) = \sin(\pi \cos \theta)$$

$$\Rightarrow u_r = \cos(\pi \cos \theta) \times (\text{cancel } -\sin) \cos \theta$$

$$= \cos(\pi \cos \theta) \times \cos \theta$$

$$u_\theta = \cos(\pi \cos \theta) \times (-\pi \sin \theta)$$

$$= (-\pi \sin \theta) \times \cos(\pi \cos \theta)$$

Ans:-

Find the indicated Partial derivation:

$$\textcircled{41} f(s, t) = te^{st} \quad , \quad f(0, 1)$$

$$\Rightarrow Rf(s, t) = f \times \frac{-s \times e^{st}}{+2} \times 1 + e^{st} \times 1$$

$$= \frac{-s t \times e^{st}}{2} + e^{st}$$

$$= 0 + e^{\frac{0}{1}}$$

$$= 1$$

$$\left. \begin{aligned} \frac{d}{dt}(e^{mt}) &= m e^{mt} \\ \frac{d}{dt}(e^{st}) &= e^{st} \times \frac{d}{dt}(s t) \\ &= s e^{st} \times 1 \\ &= s e^{st} \end{aligned} \right\}$$

42) $f(xy) = y \sin^{-1}(xy)$

$\Rightarrow f_y(xy) = y \times \frac{1}{\sqrt{1-(xy)^2}} \times x + \sin^{-1}(xy) \times 1$

$= \frac{xy}{\sqrt{1-(xy)^2}} + \sin^{-1}(xy)$

$= \frac{1 \times \frac{1}{2}}{\sqrt{1-(1 \times \frac{1}{2})^2}} + \sin^{-1}(1 \times \frac{1}{2})$

$= \frac{1}{\sqrt{3}} + \frac{\pi}{6}$

Ans:-

Page:- 025:-

Verify Clairaut's Theorem

59) $u = x^4 y^3 - y^4$

$\Rightarrow u_x = 4x^3 y^3 - 0$
 $= 4x^3 y^3$

$u_y = 3x^4 y^2 - 4y^3$

$u_{xy} = (u_x)_y$
 $= 12x^3 y^2$

$u_{yx} = (u_y)_x$
 $= 12x^3 y^2$

As $u_{xy} = u_{yx}$
(Verified)

60) $u = e^{ny} \cdot \sin x$

$$\Rightarrow u_x = \sin x (y e^{ny})$$

$$u_{xy} = (u_x)_y$$

$$= y e^{ny} \cos x + \sin x (y n e^{ny} + e^{ny} \cdot 1)$$

$$u_y = n e^{ny} \sin x + e^{ny} \cos x$$

$$(u_y)_x = \sin x (n y e^{ny} + e^{ny} \cdot 1) + y e^{ny} \cos x$$

Now

$$u_{xy} = u_{yx}$$

So. (Verified)

Find the indicated Partial derivatives :

(63) $f(x, y) = x^4 y^2 - x^3 y$; f_{xxx} , f_{xyx} , f_{xxy}

$\Rightarrow f_x = 4x^3 y^2 - 3x^2 y$; $f_y = x^4 \cdot 2y - x^3$

$(f_x)_x = 12x^2 y^2 - 6xy$

$(f_y)_y = 2x^4 - x^3$

$(f_{xx})_x = 24xy^2 - 6y$

$(f_{yy})_y = 0$

$\Rightarrow f_x = 4x^3 y^2 - 3x^2 y$

$(f_x)_y = 8x^3 y - 3x^2$

$(f_{xy})_x = 24x^2 y - 6x$

(64) $f(x, y, z) = e^{xyz^2}$; f_{xyx}

$\Rightarrow f_x = y z^2 e^{xyz^2}$

$(f_x)_y = z^2 (y \cdot x \cdot z^2 e^{xyz^2} + e^{xyz^2} \cdot 1)$
 $= z^4 y x e^{xyz^2} + z^2 e^{xyz^2}$

$\Rightarrow (f_{xy})_x = x y (z^4 \cdot 2z \cdot e^{xyz^2} + e^{xyz^2} \cdot 4z^3) + z^2 \cdot 2z e^{xyz^2} + e^{xyz^2} \cdot 2z$
 $= 2x y z^5 e^{xyz^2} + 4x y z^3 e^{xyz^2} + 2z^3 e^{xyz^2} + 2z e^{xyz^2}$

Verify Laplace's equation:

Ques -
1. $u_x + u_y = 0$

76 (a) $u = x^2 + y^2$

$\Rightarrow u_x = 2x$

$u_{xx} = (u_x)_x$
 $= 2 \quad \text{--- (1)}$

$\Rightarrow u_y = 0 + 2y$

$u_{yy} = (u_y)_y$
 $= 2 \quad \text{--- (2)}$

As (1) + (2) $\Rightarrow$

$u_{xx} + u_{yy} = 2 + 2 = 4 \neq 0$

(not verified)

(b) $u = e^{-x} \cos y - e^{-y} \cos x$

$u_x = -e^{-x} \cos y - e^{-y} (-\sin x)$
 $= -e^{-x} \cos y + e^{-y} \sin x$

$u_{xx} = (u_x)_x$

$= -(-e^{-x} \cos y) + (e^{-y} \sin x)$

$= e^{-x} \cos y + e^{-y} \sin x \quad \text{--- (i)}$

$$u_y = e^{-n} \sin y - (-e^{-y} \cos n)$$

$$= -e^{-n} \sin y + e^{-y} \cos n$$

$$(u_y)_y = -e^{-n} \cos y + (-e^{-y} \cos n)$$

$$= -e^{-n} \cos y - e^{-y} \cos n \quad \text{--- (ii)}$$

$$\text{As (i) + (ii)} \Rightarrow u_{xx} + u_{yy}$$

$$= e^{-n} \cos y + e^{-y} \cos n + (-e^{-n} \cos y - e^{-y} \cos n)$$

$$= e^{-n} \cancel{\cos y} + e^{-y} \cancel{\cos n} - e^{-n} \cancel{\cos y} - e^{-y} \cancel{\cos n}$$

$$= 0$$

(Verified)

Wave Equation:

$$\textcircled{78} \textcircled{D} u = \sin(n-at) + \ln(m+at)$$

$$\Rightarrow u_x = \cos(n-at) \cdot 1 + \frac{1}{m+at} \cdot 1$$

$$= \cos(n-at) + \frac{1}{m+at}$$

$$(u_x)_x = -\sin(n-at) \cdot 1 + \frac{-1}{(m+at)^2}$$

$$= -\sin(n-at) - \frac{1}{(m+at)^2}$$

$$\text{--- (i)}$$

$$\begin{aligned} & \frac{1}{2} n(1 + (m+at)^{-1}) \\ & = - (m+at)^{-2} \cdot 1 \\ & = - \frac{1}{(m+at)^2} \end{aligned}$$

$$u_t = \cos(m-at) \times (-a) + \frac{1}{m+at} \times a$$

$$= \cos(m-at) \times (-a) + \frac{a}{m+at}$$

$$= -\frac{a \cos(m-at)}{1} + \frac{a}{m+at}$$

$$(u_t)_t = - \left\{ a \times \sin(m-at) \times (-a) \right\} + \frac{-a^2}{(m+at)^2} = \left\{ \frac{d}{dt} \left\{ a(m+at)^{-1} \right\} \right\} = -a \times (m+at)^{-2} \times a = -\frac{a^2}{(m+at)^2}$$

$$= -a^2 \sin(m-at) - \frac{a^2}{(m+at)^2}$$

$$= a^2 \left(-\sin(m-at) - \frac{1}{(m+at)^2} \right) \quad (1)$$

from (1) $\Rightarrow$

$$u_{tt} = a^2 \sin(m-at) - \frac{a^2}{(m+at)^2}$$

(Verified)

Law:- # Maximum & Minimum Values

1. for stationary point or critical point:-

* if $f(x, y)$, then $f_x = 0$, $f_y = 0$

2. 2nd derivative test:-

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

$$D = f_{xx}f_{yy} - [f_{xy}]^2$$

$$D = f_{xx}(a, b) \times f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

1. if $D > 0$ & $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
2. if $D > 0$ & $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
3. if $D < 0$ then $f(a, b)$ is not a local maximum or minimum $f(a, b)$ has a saddle point.

4. if $D = 0$ then no

conclusion can be drawn.

P-068

(11)

$$f(x, y) = x^3 - 3x + 3xy^2$$

⇒ for stationary / critical point: —

$$f_x = 0$$

$$3x^2 - 3 + 3y^2 = 0$$

$$3(x^2 + y^2 - 1) = 0$$

$$x^2 + y^2 = 1$$

$$f_y = 0$$

$$2 \times 3xy = 0$$

$$6xy = 0$$

$$x = 0, y = 0$$

for $x = 0, y^2 = 1 \Rightarrow y = \pm 1$

for $y = 0, x^2 = 1 \Rightarrow x = \pm 1$

∴ Stationary point = $(0, 1), (0, -1), (1, 0), (-1, 0)$

⇒ for local maximum or minimum point: —

$$f_x = 3x^2 + 3y^2 - 3$$

$$f_{xx} = 6x + 0 - 0$$

$$= 6x$$

$$f_{yy} = 6x$$

$$f_{xy} = 6y$$

$$D(x, y) = f_{xx} f_{yy} - [f_{xy}]^2$$

$$= 6x \times 6y - (6y)^2$$

$$= 36y^2 - 36y^2$$

Stationary point	$D(a,b)$	$f_{xx}(a,b)$	Result
$(0,1)$	$-36 < 0$		Saddle point
$(0,-1)$	$-36 < 0$		Saddle point
$(1,0)$	$36 > 0$	$6 > 0$	local minimum
$(-1,0)$	$36 > 0$	$-6 < 0$	local maximum

So, Maximum Value $= x^3 - 3x + 3xy^2$

$$= (-1)^3 - (3 \times -1) + 3 \times (-1) \times (0)^2$$

$$= 2$$

Minimum Value $= x^3 - 3x + 3xy^2$

$$= (1)^3 - 3 \times 1 + 3 \times 1 \times 0^2$$

$$= -2$$

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$$f(x, y) = x^2 + y^4 + 2xy$$

for: Stationary points

$$\Rightarrow f_x = 0$$

$$\Rightarrow 2x + 2y = 0$$

$$\Rightarrow x + y = 0$$

$$\Rightarrow x = -y \quad \text{--- (1)}$$

$$\Rightarrow f_y = 0$$

$$\Rightarrow 4y^3 + 2x = 0$$

$$\Rightarrow 2(2y^3 + x) = 0$$

$$\Rightarrow 2y^3 + x = 0$$

$$\Rightarrow 2y^3 + (-y) = 0$$

$$\Rightarrow 2y^3 - y = 0$$

$$\Rightarrow y(2y^2 - 1) = 0$$

$$\text{So, } y = 0, 2y^2 = 1$$

$$y = \pm \frac{1}{\sqrt{2}}$$

$$\text{or, } y = \pm \frac{1}{\sqrt{2}}$$

$$\text{for } y = 0 \Rightarrow x = 0$$

$$y = +\frac{1}{\sqrt{2}} \Rightarrow x = -\frac{1}{\sqrt{2}}$$

$$y = -\frac{1}{\sqrt{2}} \Rightarrow x = +\frac{1}{\sqrt{2}}$$

So, the Stationary point is $(0, 0), (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

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for local maximum or minimum:—

$$f_{xx} = 2$$

$$f_{xy} = 12xy^2$$

$$f_{yy} = 2$$

$$D = f_{xx}(a,b) \times f_{yy}(a,b) - [f_{xy}(a,b)]^2$$

$$= 2 \times 12x^2 - 4$$

$$= 24x^2 - 4$$

Stationary point	$D(x,y)$	$f_{xx}(x,y)$	Conclusions
$(0,0)$	$-4 < 0$		Saddle point
$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	$8 > 0$	$2 > 0$	Local minimum
$(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	$8 > 0$	$2 > 0$	Local minimum

So, minimum Value is $= x^2 + y^4 + 2xy$

$$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^4 + 2 \times \frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}}$$

$$= -\frac{1}{4}$$

③ $f(x,y) = 4 + x^2 + y^3 - 3xy$

for Stationary points—

$$f_x = 0$$

$$\Rightarrow 2x^2 - 3y = 0$$

$$\Rightarrow x^2 = \frac{3}{2}y \quad \text{--- (I)}$$

$$\Rightarrow (y^2)^2 = y \quad \text{from (I) } \Rightarrow$$

$$f_y = 0$$

$$\Rightarrow 3y^2 - 3x = 0$$

$$\Rightarrow y^2 = x \quad \text{--- (II)}$$

$$\Rightarrow y^4 = y$$

$$\Rightarrow y^4 - y = 0$$

$$\Rightarrow y(y^3 - 1) = 0$$

$$\underline{\text{So,}} \quad y = 0, \quad y^3 - 1 = 0$$

$$(y-1)(y^2 + y + 1) = 0$$

$$y = 1$$

$$y = (0, 1)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$(y^3 - 1) = (y-1)(y^2 + y + 1)$$

now,

$$y = 0 \Rightarrow n = 0 \text{ for (i)}$$

$$y = 1 \Rightarrow n = \pm 1$$

Again

$$y = 0 \Rightarrow n = 0$$

$$y = 1 \Rightarrow n = \pm 1 \text{ for (ii)}$$

So Stationary point is $(0, 0), (1, 1), (-1, 1)$

for local maximum or minimum

$$f_{xx} = 6x$$

$$f_{yy} = 6y$$

$$f_{xy} = -3$$

$$D = f_{xx}f_{yy} - [f_{xy}]^2$$

$$= 6x \times 6y - (-3)^2$$

$$= (36xy - 9)$$

Stationary point	$f_{xx}(a,b)$	$D(a,b)$	Result
$(0,0)$		$-9 < 0$	Saddle point
$(-1,1)$		$-45 < 0$	Saddle point
$(1,1)$	$6 > 0$	$27 > 0$	Local minimum

So, the minimum Value is: $4 + x^3 + y^3 - 3xy$
 $= 4 + (1)^3 + (1)^3 - 3 \times 1 \times 1$
 $= 2$

Taylor polynomial for multiple Variable function:-

1. 2nd degree Taylor polynomial of $f(x,y)$ near the point (a,b) is:-

$$\Rightarrow f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) + \frac{f_{xx}(a,b)(x-a)^2}{2!} + \frac{f_{yy}(a,b)(y-b)^2}{2!} + f_{xy}(a,b)(x-a)(y-b) + \dots$$

Example:-

⇒ Determine the 1st and 2nd degree Taylor polynomial approximations for the following function $f(x,y)$ of x and y near the point.

$$f(x,y) = \sin 2x + \cos y \quad (0,0)$$

$$\Rightarrow f(a,b) = \sin 2 \times 0 + \cos 0 = 1$$

$$\Rightarrow f_x(a,b) = \cos 2x \times 2 \\ = 2 \times \cos 2 \times 0 = 2$$

$$\Rightarrow f_y(a,b) = -\sin y = 0$$

$$\Rightarrow f_{xx}(a,b) = 2 \times (-\sin 2x) \times 2 \\ = -4 \sin 2x = 0$$

$$\Rightarrow f_{yy}(a,b) = +(\cos y) \\ = -\cos y = -1$$

$$\Rightarrow f_{xy}(a,b) = 0$$

So, The Taylor polynomial for the function is:-

$$f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) + \frac{f_{xx}(a,b)(x-a)^2}{2} \\ + \frac{f_{yy}(a,b)(y-b)^2}{2} + f_{xy}(a,b)(x-a)(y-b) + \dots$$

$$= 1 + 2(x-a) + 0 + 0 + \frac{-1}{2}(y-b)^2 + 0 \\ = 1 + 2(x-a) - \frac{1}{2}(y-b)^2$$

$$= 1 + 2(x-0) + \frac{-1}{2}(y-0)^2$$

$$= 1 + 2x - \frac{1}{2}y^2$$

Ans

* $f(x,y) = xe^y + 1$ at $(1,0)$

$$\Rightarrow f(a,b) = 1e^0 + 1 = 2$$

$$\Rightarrow f_x(a,b) = e^y = 1$$

$$\Rightarrow f_y(a,b) = xe^y = 1$$

$$\Rightarrow f_{xx}(a,b) = 0$$

$$\Rightarrow f_{xy}(a,b) = xe^y = 1$$

$$\Rightarrow f_{yy}(a,b) = e^y = 1$$

So, the Taylor polynomial for the function is —

$$f(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) + \frac{f_{xx}(a,b)(x-a)^2}{2} + \frac{f_{yy}(a,b)(y-b)^2}{2} + f_{xy}(a,b)(x-a)(y-b)$$

$$= 2 + (x-1) + (y-0) + \frac{0 \times (x-1)^2}{2} + \frac{(y-0)^2}{2} + (x-1)(y-0)$$

$$= 2 + (x-1) + y + \frac{y^2}{2} + y(x-1)$$

Ans

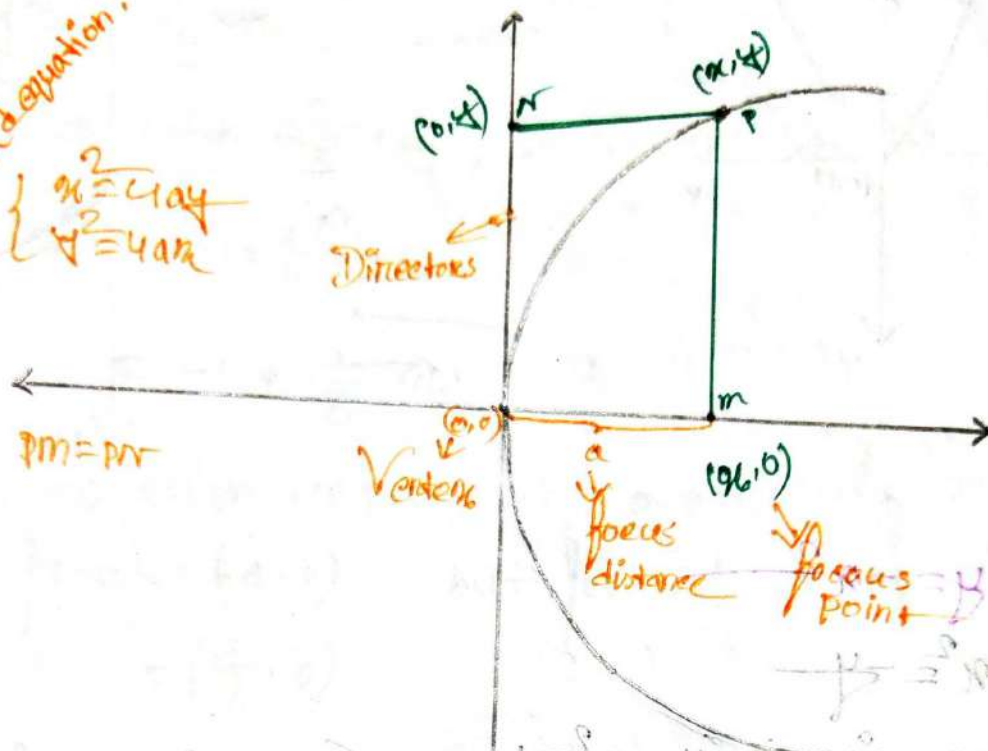
Coordinate Geometry of Two Dimensions.

Conic Sections

Parabola

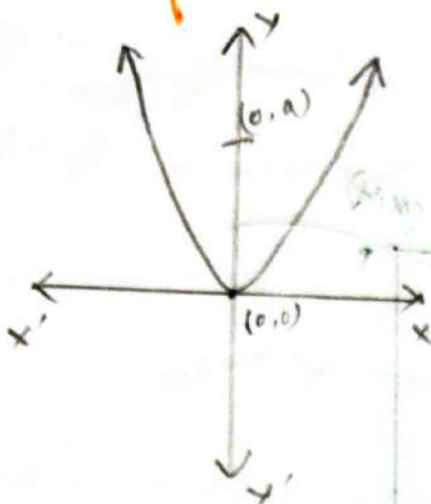
Standard equation

$$\begin{cases} x^2 = 4ay \\ y^2 = 4ax \end{cases}$$

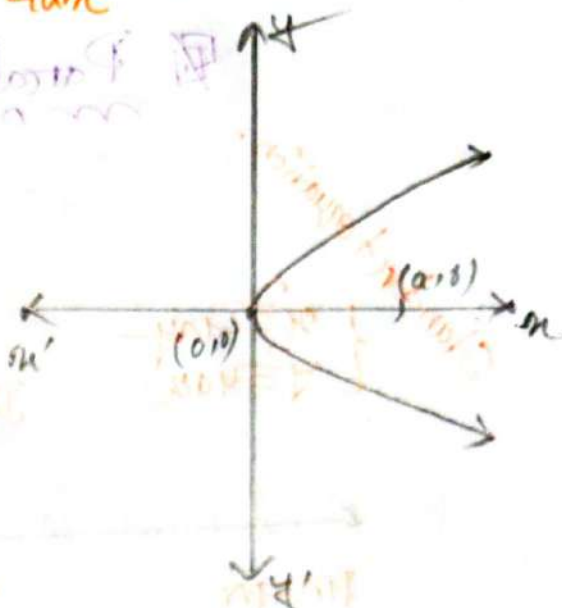


Equation	Vertex	Axis along	Distance of the focus from the vertex
$x^2 = 4ay$	$(0,0)$	y-axis	$(0,a)$
$y^2 = 4ax$	$(0,0)$	x-axis	$(a,0)$
$(x-h)^2 = 4a(y-k)$	(h,k)	y-axis	$(h, k \pm a)$
$(y-k)^2 = 4a(x-h)$	(h,k)	x-axis	$(h \pm a, k)$

$$x^2 = 4ay$$



$$y^2 = 4ax$$



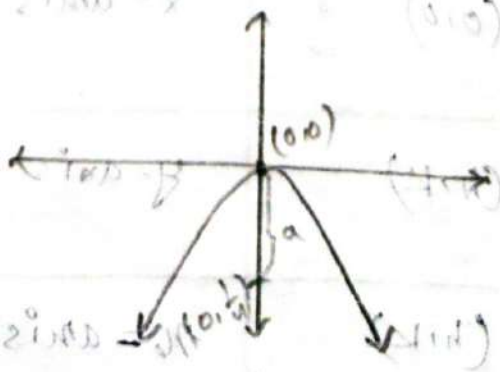
(11) $y = -x^2$

$$\Rightarrow x^2 = -y$$

Comparing with $x^2 = 4ay$, So, we can write.

$$x^2 = -4 \times \frac{1}{4} \times y$$

The equation represent a parabola with vertex (0,0),
And focus = (0, -a) but focus distance = (0, a)
= (0, -\frac{1}{4}) = (0, \frac{1}{4})



$$(14) \quad m = -2y^2$$

$$\Rightarrow y^2 = -\frac{m}{2}$$

$$\Rightarrow y^2 = -\frac{1}{2} \times m$$

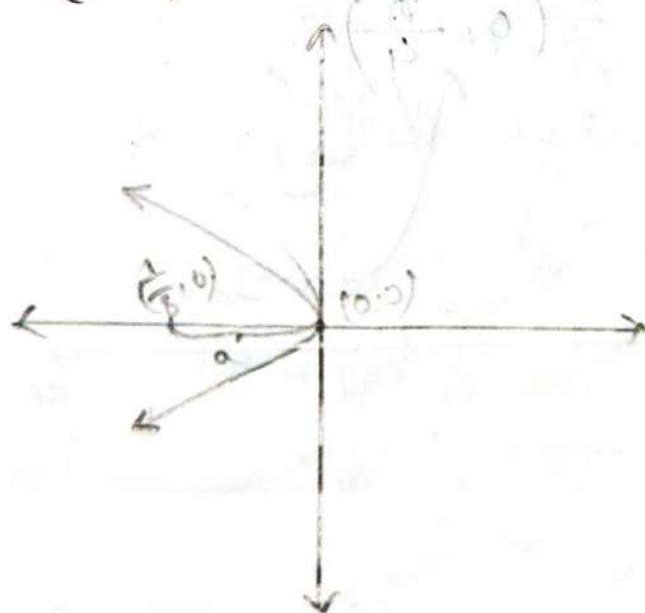
comparing with $m^2 = 4ay$, so, we can write

$$y^2 = -4 \times \frac{1}{4} \times \frac{1}{2} m$$

$$= -4 \times \frac{1}{8} m$$

The equation represent a parabola with vertex (0,0)
 And focus = (-a, 0) but focus distance = (-a, 0)

$$= \left(-\frac{1}{8}, 0\right) \quad \left(-\frac{1}{8}, 0\right) = \left(-\frac{1}{8}, 0\right)$$



$$(18) \quad y = x^2 + 2$$

$$\Rightarrow x^2 = y - 2$$

Comparing with $(x-h)^2 = 4a(y-k)$,

So, we can write.

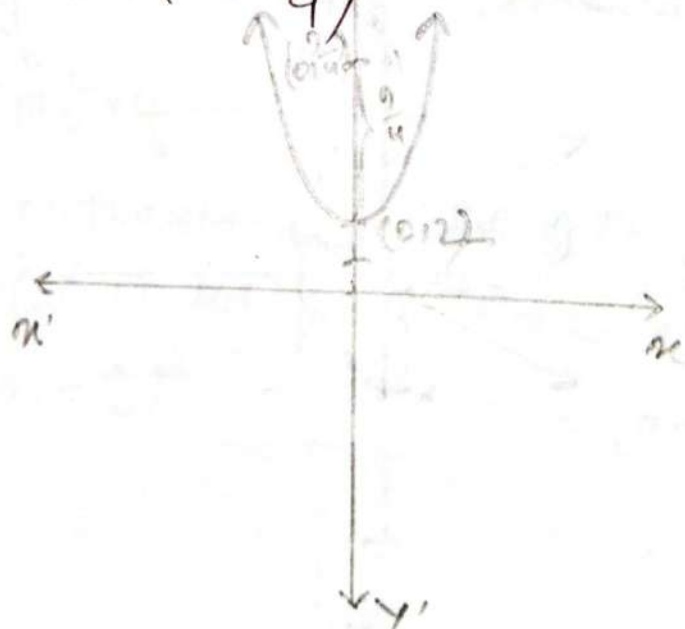
$$(x-0)^2 = 4 \times \frac{1}{4} (y+2)$$

The following equation represent a parabola with Vertex $(0, 2)$.

And the focus = $(h, k+a)$

$$= (0, 2 + \frac{1}{4})$$

$$= (0, \frac{9}{4})$$



$$(12) \quad x_0 = y^2 - 1$$

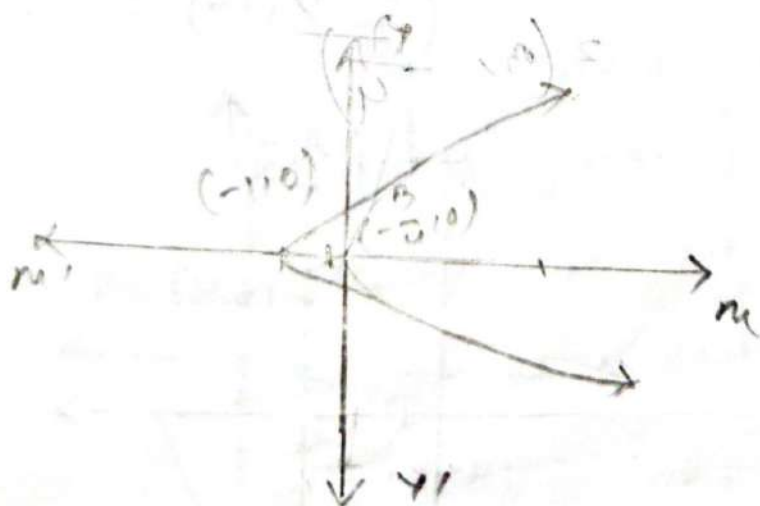
$$\Rightarrow y^2 = x + 1$$

Compare with $(y-k)^2 = 4ax(h-k)$
So, we can write.

$$(y-0)^2 = 4 \times \frac{1}{4} (x-(-1))$$

The following equation represent a parabola with
 vertex $(-1, 0)$,

And the focus $= (h+a, k)$
 $= (-1 + \frac{1}{4}, 0)$
 $= (-\frac{3}{4}, 0)$



*27 $y = x^2 - 6x + 13$

$\Rightarrow y = x^2 - 3 \cdot x \cdot 2 + (3)^2 + 4$

$\Rightarrow y = (x-3)^2 + 4$

$\Rightarrow (x-3)^2 = y-4$

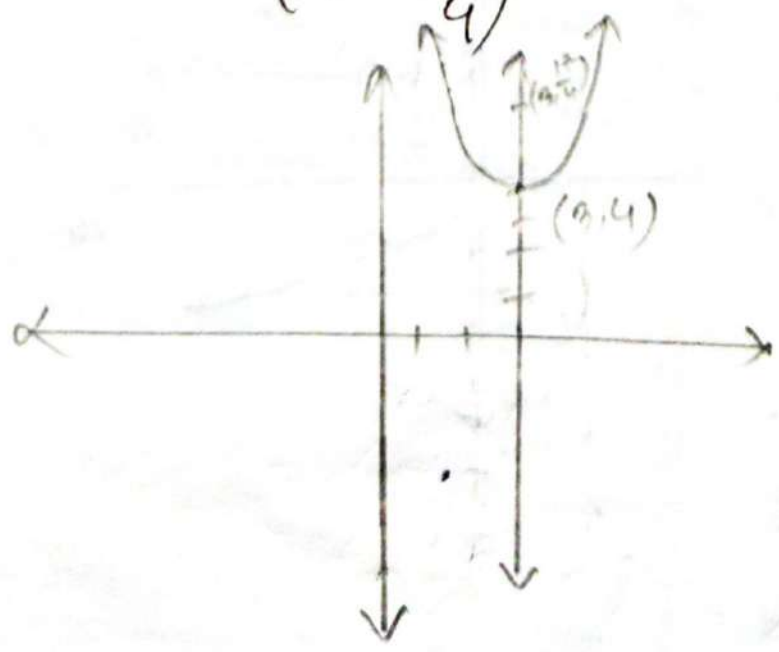
Compare with $(x-h)^2 = 4a(y-k)$

So, we can write,

$(x+3)^2 = 4 \times \frac{1}{4} (y+4)$

So, the following function represent a parabola with vertex $(h, k) = (3, 4)$

And, the focus $= (h, k+a)$
 $= (3, 4 + \frac{1}{4})$
 $= (3, \frac{17}{4})$



(24) $y = x^2 + 2x$

$$\Rightarrow y = x^2 + 2x + 1 - 1$$

$$\Rightarrow y = (x+1)^2 - 1$$

$$\Rightarrow (x+1)^2 = y+1$$

Compare with $(x-h)^2 = 4a(y-k)$

So, we can write,

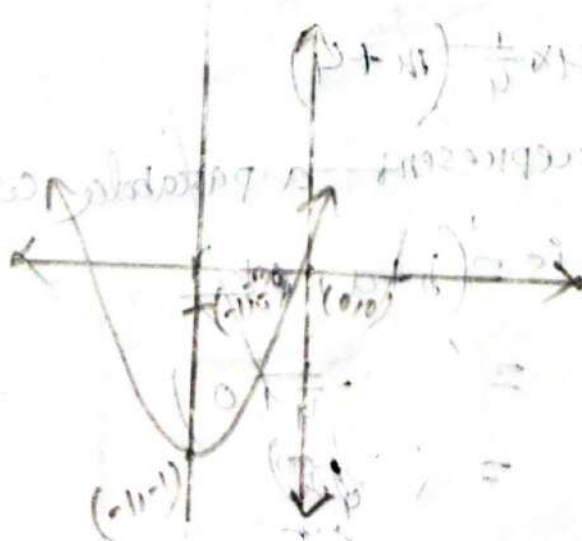
$$(x+1)^2 = 4 \times \frac{1}{4} (y+1)$$

So, the following function represent a parabola with vertex $(h, k) = (-1, -1)$

And the focus $= (h, k+a)$

$$= (-1, -1 + \frac{1}{4})$$

$$= (-1, -\frac{3}{4})$$



③② $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ $x^2 + 2x + 1 = (x+1)^2$ (12)

$\Rightarrow$ $4x \Rightarrow 8x - 16 = 0$

$\Rightarrow x = 2$

$9y \Rightarrow 18y + 54 = 0$

$\Rightarrow y = -3$

So, the vertex of the following equation is $(2, -3)$ and it represents an ellipse.

②② $x = 4 - y^2$

$\Rightarrow y^2 = 4 - x$

$\Rightarrow y^2 = -x + 4 \Rightarrow y^2 = -1(x - 4)$

Compare with $(y - k)^2 = 4a(x - h)$

So, we can write,

$$(y - 0)^2 = -4 \times \frac{1}{4} (x - 4)$$

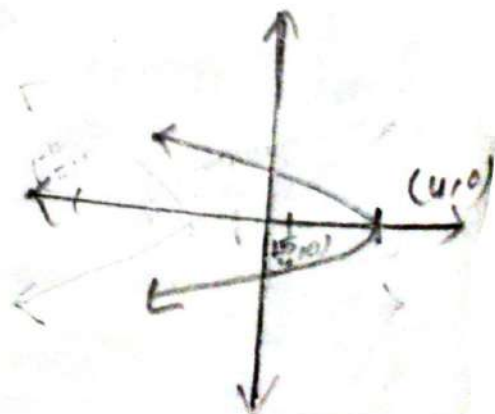
The equation represents a parabola with vertex $(h, k) = (4, 0)$

And the focus is $(h + a, k)$

$= (4 - \frac{1}{4}, 0)$

$= (\frac{15}{4}, 0)$

Ans

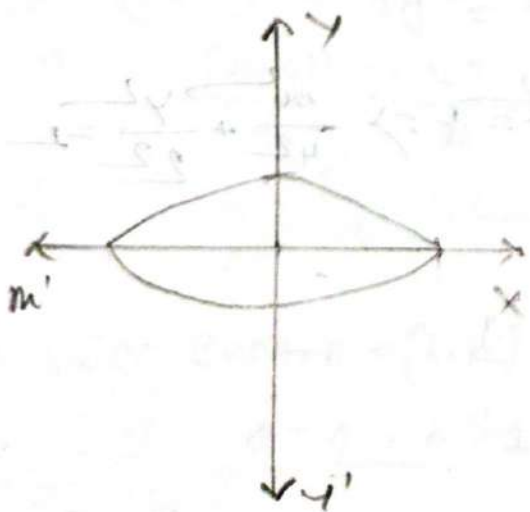


Ellipse

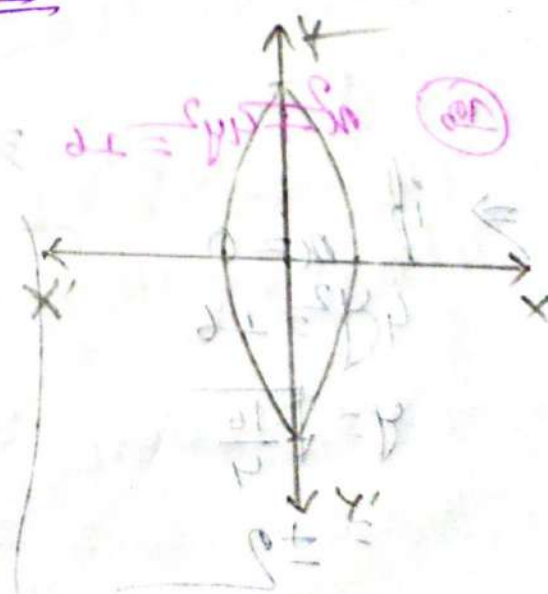
⇒ An ellipse is the set of all points in a plane the sum of whose distances from two fixed points, called foci, is a constant.

Equation	Dist. of the focus from the center	Center	Vertices	Major Axis along	minor axis
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$	$c = \sqrt{a^2 - b^2}$	$(0, 0)$	$(\pm a, 0)$	x-axis	y-axis
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a < b$	$c = \sqrt{b^2 - a^2}$	$(0, 0)$	$(0, \pm b)$	y-axis	x-axis
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$		(h, k)	$(h, k \pm b)$ Vertices $(h \pm a, k)$	y-axis	x-axis

if $a > b$:-



if $a < b$:-



$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{5^2} + \frac{y^2}{4^2} = 1$$

if $y=0$
then, $\frac{x^2}{25} = 1$

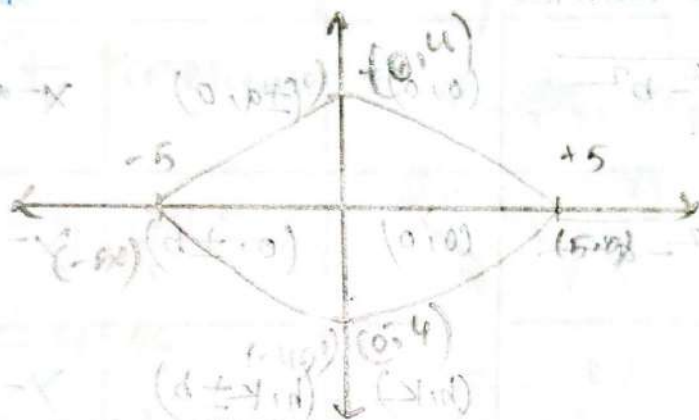
$$x = \sqrt{25}$$

$$= \pm 5$$

if $x=0$
then $\frac{y^2}{16} = 1$

$$y = \sqrt{16}$$

$$= \pm 4$$



here, major axis = x axis

So, the Vertices are $(\pm a, 0)$
 $= (\pm 5, 0)$

$$\textcircled{10} \quad x^2 + 4y^2 = 16 \Rightarrow \frac{x^2}{16} + \frac{4y^2}{16} = 1 \Rightarrow \frac{x^2}{4^2} + \frac{y^2}{2^2} = 1$$

$\Rightarrow$ if $x=0$

$$4y^2 = 16$$

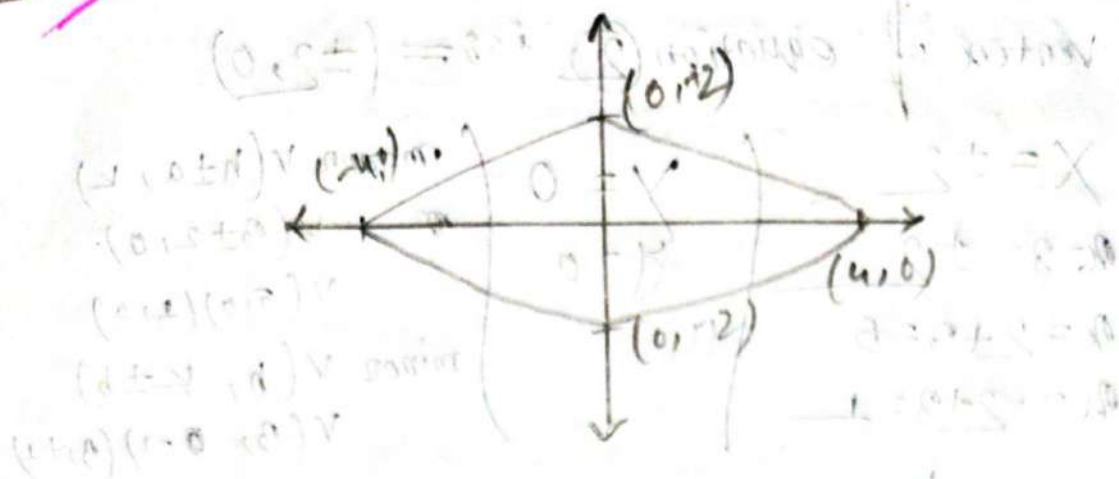
$$y = \sqrt{\frac{16}{4}}$$

$$= \pm 2$$

if $y=0$

$$x^2 = 16$$

$$x = \pm 4$$



here, major axis = x axis

So, the vertices are $(\pm a, 0)$
 $= (\pm 4, 0)$

31. $x^2 + 4y^2 - 6x + 5 = 0$

$$\Rightarrow x^2 - 2 \times x \times 3 + (3)^2 + 4y^2 + 5 - 9 = 0$$

$$\Rightarrow (x-3)^2 + 4y^2 - 4 = 0$$

$$\Rightarrow (x-3)^2 + 4y^2 = 4$$

$$\Rightarrow \frac{(x-3)^2}{4} + \frac{y^2}{1} = 1$$

$$\Rightarrow \frac{(x-3)^2}{2^2} + \frac{y^2}{1^2} = 1 \quad \text{--- (i)}$$

So, center = $(h, k) = (3, 0)$

$$a = 2, b = 1$$

Let, $X = x - 3, Y = y$

So, we can write,

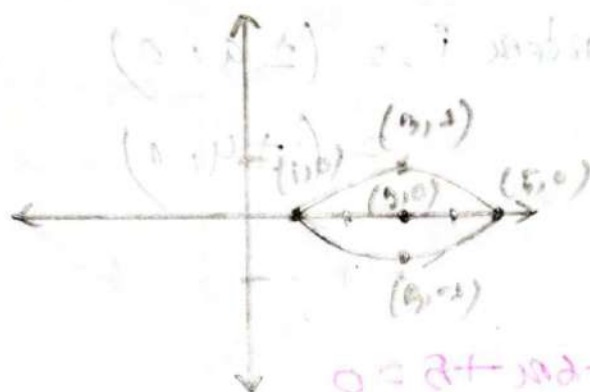
$$\frac{X^2}{2^2} + \frac{Y^2}{1^2} = 1 \quad \text{--- (ii)}$$

The major axis = x axis
 $y = \pm 1$

So, the vertices of equation (2) is $(\pm 2, 0)$

$$\left. \begin{array}{l} X = \pm 2 \\ x - 3 = \pm 2 \\ x = 2 + 3 = 5 \\ x = -2 + 3 = 1 \end{array} \right\} \begin{array}{l} y = 0 \\ y = 0 \end{array} \left(\begin{array}{l} \text{major } V(h \pm a, k) \\ V(3 \pm 2, 0) \\ V(5, 0) (1, 0) \\ \text{minor } V(h, k \pm b) \\ V(3, \pm 1) (3, +1) \end{array} \right)$$

So, vertices of equation (1) is $(5, 0) (1, 0)$



$$\begin{aligned} c &= \sqrt{a^2 - b^2} \\ &= \sqrt{3^2 - 1^2} \\ &= 2\sqrt{2} \end{aligned}$$

focus $(h \pm c, k) = (3 \pm 2\sqrt{2})$

(32) $4x^2 + 9y^2 - 16x + 54y + 61 = 0$

$$\Rightarrow 4x^2 - 16x + 9y^2 + 54y + 61 = 0$$

$$\Rightarrow 4(x^2 - 4x) + 9(y^2 + 6y) + 61 = 0$$

$$\Rightarrow 4 \left\{ x^2 - 2 \times x \times 2 + (2)^2 \right\} + 9 \left\{ y^2 + 2 \times y \times 3 + (3)^2 \right\} + 61 - 16 - 81 = 0$$

$$\Rightarrow 4(x - 2)^2 + 9(y + 3)^2 - 36 = 0$$

$$\Rightarrow 4(x - 2)^2 + 9(y + 3)^2 = 36$$

$$\Rightarrow \frac{(x - 2)^2}{9} + \frac{(y + 3)^2}{4} = 1$$

$$\Rightarrow \frac{(x-2)^2}{3^2} + \frac{(y+3)^2}{2^2} = 1 \quad (1)$$

center is $(2, -3)$

$$a=3, b=2$$

major axis vertices is $(h \pm a, k)$

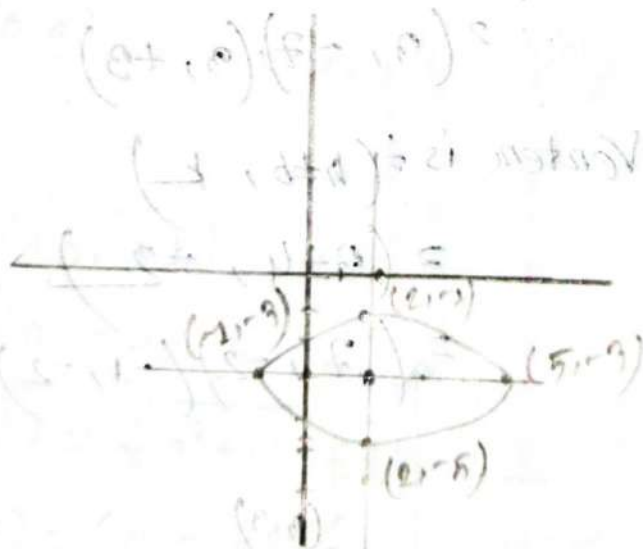
$$= (2 \pm 3, -3)$$

$$= (5, -3) \text{ and } (-1, -3)$$

minor axis vertices is $(h, k \pm b)$

$$= (2, -3 \pm 2)$$

$$= (2, -5) \text{ and } (2, -1)$$



$$e = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\sqrt{9 - 4}}{3} = \frac{\sqrt{5}}{3}$$

$$\sqrt{5}$$

$$\text{foci is } (h \pm e, k)$$

$$= (2 \pm \frac{\sqrt{5}}{3}, -3)$$

Ans

Ans

$$25(x-3)^2 + 16(y+2)^2 = 400$$

$$\Rightarrow \frac{25(x-3)^2}{400} + \frac{16(y+2)^2}{400} = 1$$

$$\Rightarrow \frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1$$

$$\Rightarrow \frac{(x-3)^2}{4^2} + \frac{(y+2)^2}{5^2} = 1$$

$\Rightarrow$ So, center is $(h, k) = (3, -2)$

$$a = 5, b = 4$$

major axis Vertices is $(h, k \pm a)$

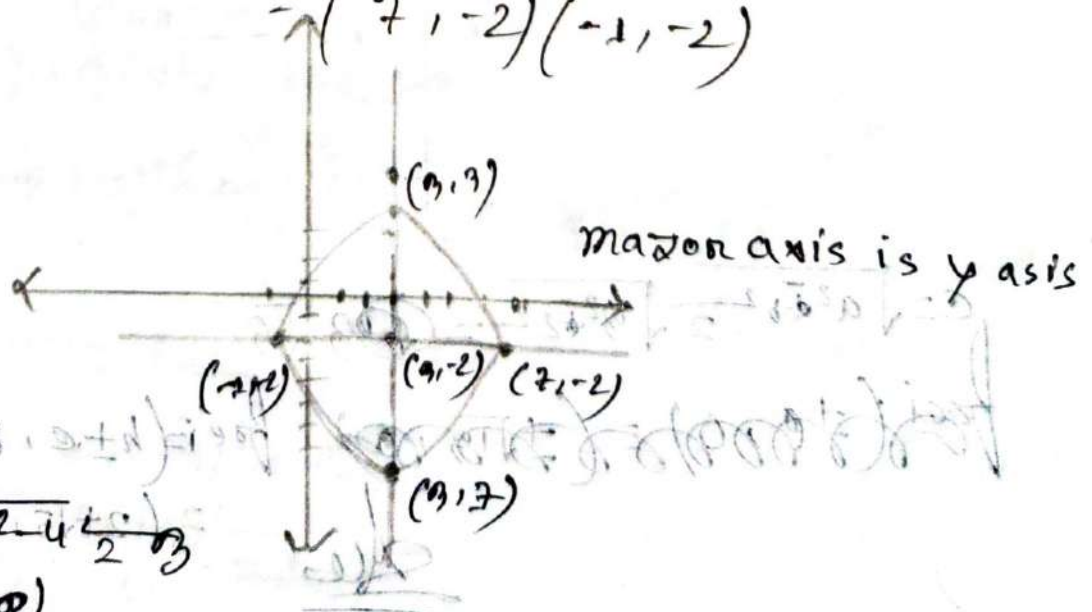
$$= (3, -2 \pm 5)$$

$$= (3, -7), (3, +3)$$

minor axis Vertices is $(h \pm b, k)$

$$= (3 \pm 4, -2)$$

$$= (7, -2), (-1, -2)$$



$$c = \sqrt{a^2 - b^2} = \sqrt{5^2 - 4^2} = 3$$

$$\text{foci} = (h, k \pm c)$$

$$= (3, -2 \pm 3)$$

29/1/20
M3: Lec 2

Hyperbolas

Equation	Distance of the all foci from the center	Center	Symptotes
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$c = \sqrt{a^2 + b^2}$	$(0, 0)$	$y = \pm \frac{bx}{a}$
$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$	$c = \sqrt{a^2 + b^2}$	$(0, 0)$	$y = \pm \frac{bx}{a}$
$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	$c = \sqrt{a^2 + b^2}$	(h, k)	$y - k = \pm \frac{b(x-h)}{a}$

⑩ $y^2 - x^2 = 1$

$\Rightarrow \frac{y^2}{1^2} - \frac{x^2}{1^2} = 1$

This equation represents a parabola.

Center is $(0, 0)$

$a = 1$

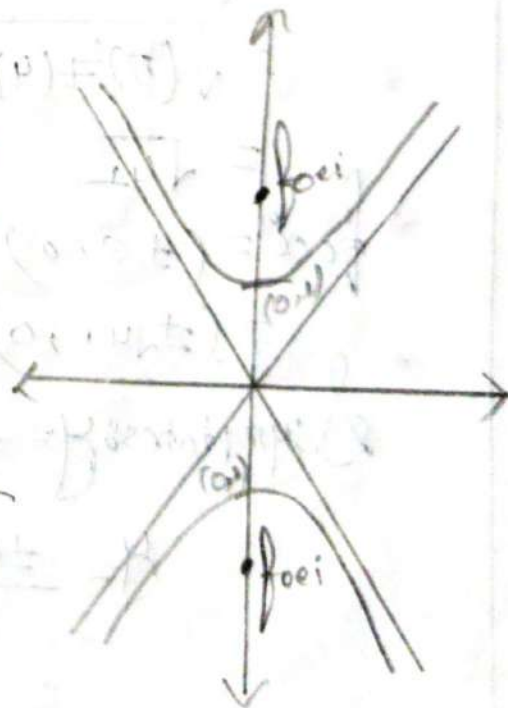
$b = 1$

Vertices $= (\pm a, \pm b)$
 $= (0, \pm 1)$

$c = \sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$

foci $= (0, \pm c) = (0, \pm \sqrt{2})$

Symptotes $y = \pm \frac{bx}{a} = \pm \frac{1x}{1}$



$$(18) \quad 16x^2 - 25y^2 = 400$$

$$\Rightarrow \frac{16x^2}{400} - \frac{25y^2}{400} = 1$$

$$\Rightarrow \frac{x^2}{25} - \frac{y^2}{16} = 1$$

$$\Rightarrow \frac{x^2}{5^2} - \frac{y^2}{4^2} = 1$$

Center of the Hyperbola = (0, 0)

$$a = 5, b = 4$$

$$\text{Vertices} = (\pm a, 0)$$

$$= (\pm 5, 0)$$

$$c = \sqrt{a^2 + b^2}$$

$$= \sqrt{(5)^2 + (4)^2}$$

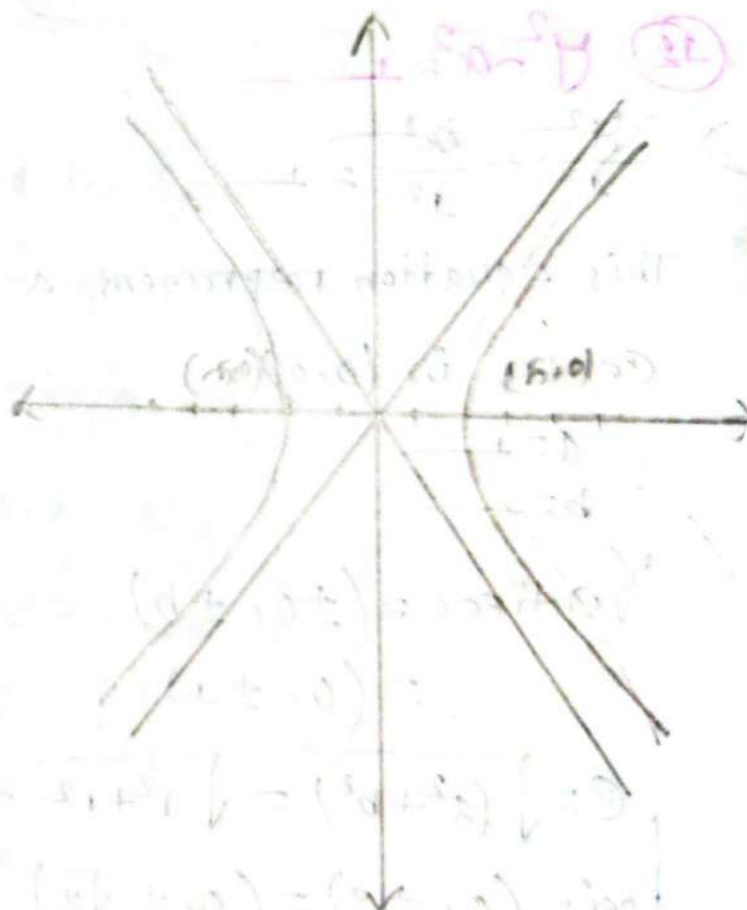
$$= \sqrt{41}$$

$$\text{foci} = (\pm c, 0)$$

$$= (\pm \sqrt{41}, 0)$$

$$\text{Symptotes } y = \frac{\pm bx}{a}$$

$$y = \frac{\pm 4x}{5}$$



$$(28) x^2 - y^2 - 4x + 3 = 0$$

$$\Rightarrow x^2 - 4x - y^2 + 3 = 0$$

$$\Rightarrow x^2 - 2 \times x \times 2 + (2)^2 - y^2 + 3 - 4 = 0$$

$$\Rightarrow (x-2)^2 - y^2 - 1 = 0$$

$$\Rightarrow (x-2)^2 - y^2 = 1$$

$$\Rightarrow \frac{(x-2)^2}{1^2} - \frac{y^2}{1^2} = 1 \quad \text{--- (1)}$$

So, Center is $(h, k) = (2, 0)$

$$a = 1$$

$$b = 1$$

$$\text{Let, } X = x - 2$$

$$Y = y$$

Compare the equation in $X^2 - Y^2 = 1$ --- (2)

So, Vertices of the equation (2) is $(\pm 1, 0)$

$$X = \pm 1 \quad ; \quad Y = 0$$

$$x - 2 = \pm 1 \quad ; \quad y = 0$$

$$x = 2 \pm 1$$

$$x = (3, 1)$$

So, Vertices of the equation (2) is

$$= (3, 0) (1, 0)$$

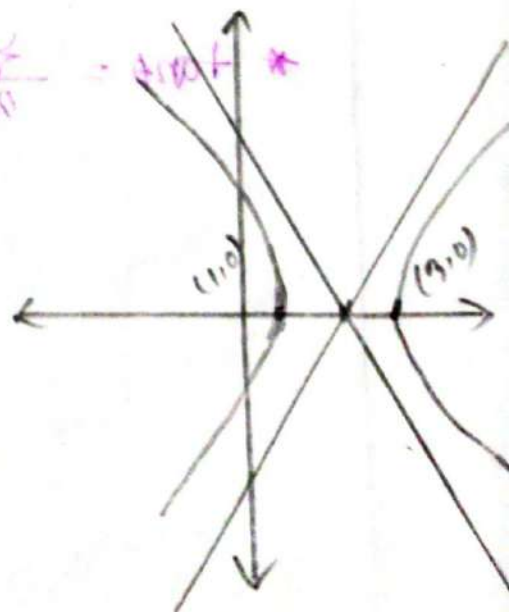
$$c = \sqrt{a^2 + b^2} = \sqrt{2}$$

$$\text{foci} = (h \pm c, k)$$

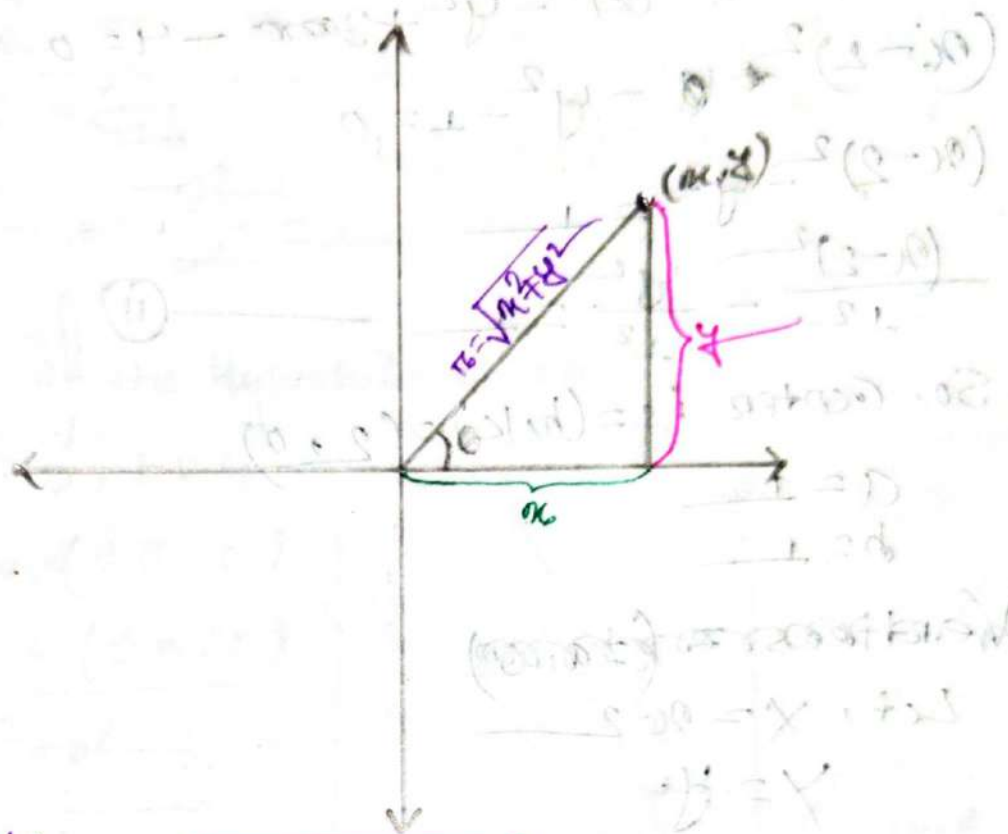
$$2 (2 \pm \sqrt{2}, 0)$$

Ans

$$\begin{aligned} y - k &= \pm \frac{b(a-h)}{a} \\ y &= \pm \frac{1(x-2)}{1} \\ y &= +(x-2) \\ y &= -(x-2) \end{aligned}$$



Relation between Polar and Cartesian Coordinate: —

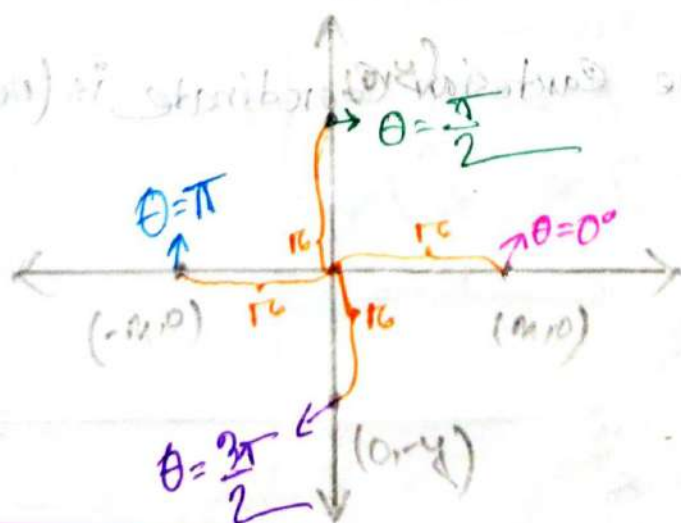
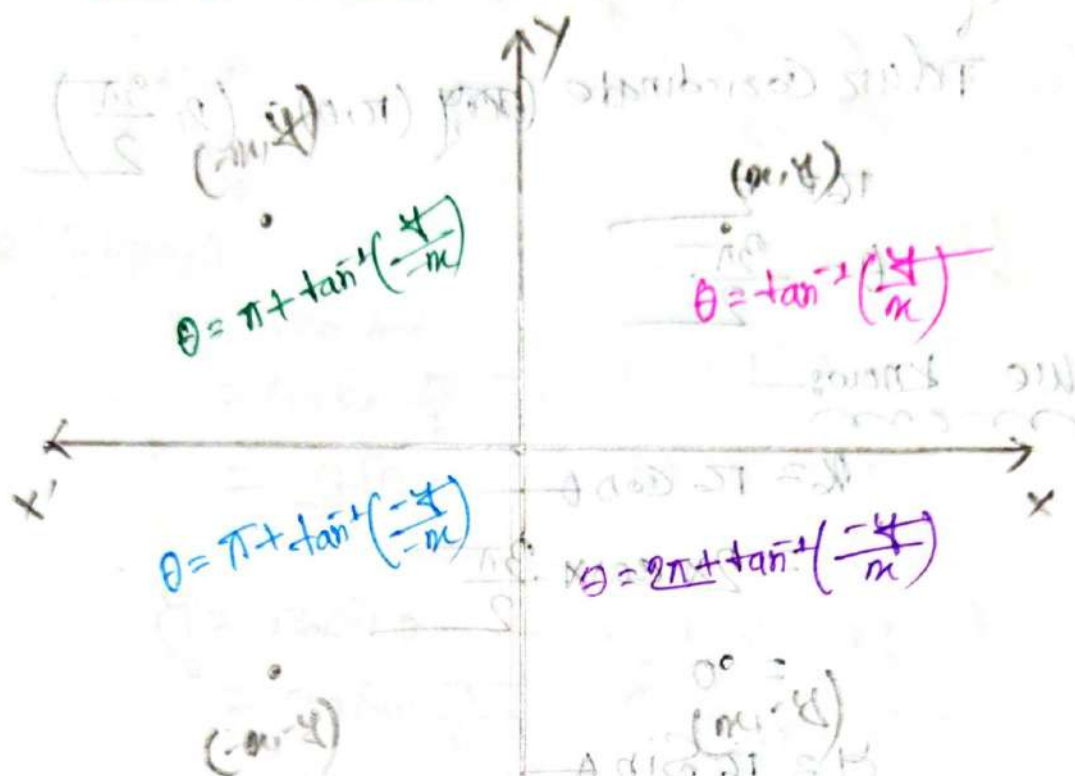


$$\star \sin \theta = \frac{y}{r} \Rightarrow \boxed{y = r \sin \theta}$$

$$\star \cos \theta = \frac{x}{r} \Rightarrow \boxed{x = r \cos \theta}$$

$$\star \tan \theta = \frac{y}{x} \Rightarrow \boxed{\theta = \tan^{-1} \left(\frac{y}{x} \right)}$$

Standard Position of Angles in the Coordinate plane:-



* Polar Coordinate (r, θ)

* Cartesian Coordinate (x, y)

Q:- Plot the point and find the Cartesian Coordinate of the point $(2, \frac{3\pi}{2})$

$\Rightarrow$ is given,

Polar Coordinate

$$(r, \theta) = (2, \frac{3\pi}{2})$$

$$r = 2$$

$$\theta = \frac{3\pi}{2}$$

$$\theta = \pi + \frac{\pi}{2} = \frac{3\pi}{2}$$

We know:-

$$x = r \cos \theta$$

$$\left(\frac{x}{r}\right) = \cos \theta \Rightarrow \left(\frac{x}{2}\right) = \cos \frac{3\pi}{2}$$

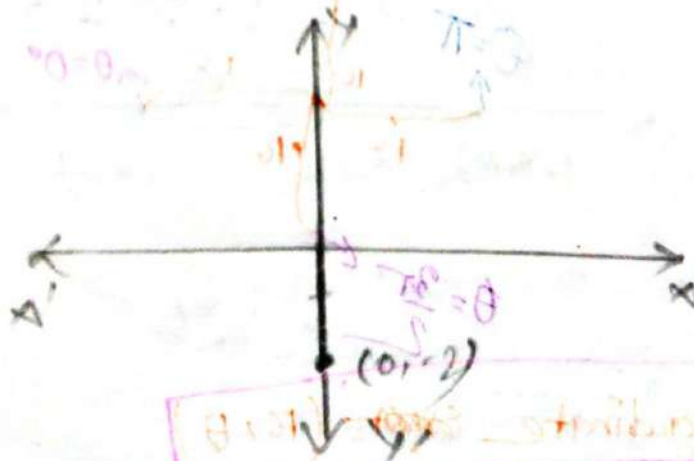
$$= 0$$

$$y = r \sin \theta$$

$$= 2 \sin \frac{3\pi}{2}$$

$$= -2$$

So, the Cartesian Coordinate is $(x, y) = (0, -2)$



$\theta = \pi + \frac{\pi}{2} = \frac{3\pi}{2}$
 $(r, \theta) = (2, \frac{3\pi}{2})$

11) find the Cartesian Coordinate of the point $(3, \frac{\pi}{6})$

is given,

$$\text{Polar Coordinate } (r, \theta) = (3, \frac{\pi}{6})$$

$$r = 3$$

$$\theta = \frac{\pi}{6}$$

we know:-

$$x = r \cos \theta$$

$$= 3 \times \cos \frac{\pi}{6}$$

$$= \frac{3\sqrt{3}}{2}$$

$$y = r \sin \theta$$

$$= 3 \times \sin \frac{\pi}{6}$$

$$= \frac{3}{2}$$

So, the Cartesian Coordinate is $(x, y) = (\frac{3\sqrt{3}}{2}, \frac{3}{2})$



Q2 Given $(\sqrt{3}, -1)$: find (r, θ)

$\Rightarrow$ is given,

the Cartesian Coordinate is $(x, y) = (\sqrt{3}, -1)$

$$x = \sqrt{3}$$

$$y = -1$$

We know,

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{(\sqrt{3})^2 + (-1)^2}$$

$$= 2$$

$$\theta = 2\pi + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$$

$$= 2\pi + \left(-\frac{\pi}{6}\right)$$

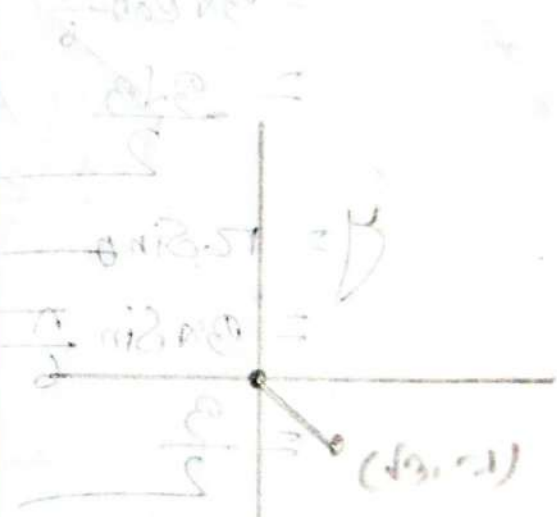
$$= 2\pi - \frac{\pi}{6}$$

$$= \frac{12\pi - \pi}{6}$$

$$= \frac{11\pi}{6}$$

$$\text{So, } (r, \theta) = \left(2, \frac{11\pi}{6}\right)$$

Ans:-



* Given $(-4, 4)$; find (r, θ) .

is given, $x = -4$
 $y = 4$

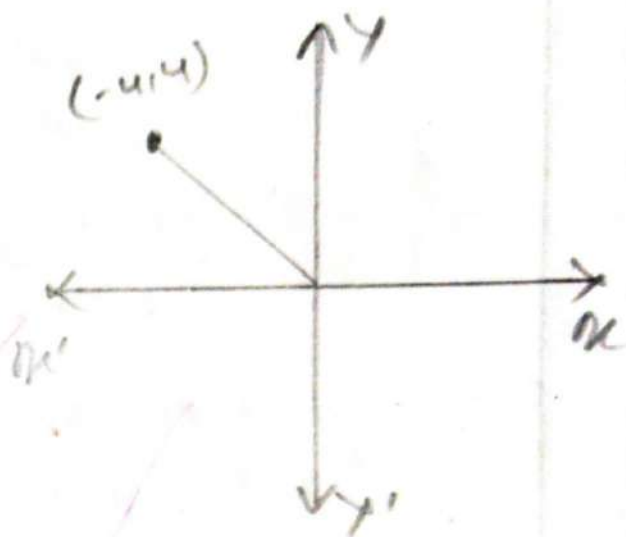
we know,
from

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-4)^2 + 4^2} \\ &= \sqrt{16 + 16} \\ &= \sqrt{32} \end{aligned}$$

$$\begin{aligned} \theta &= \pi + \tan^{-1}\left(\frac{y}{-x}\right) \\ &= \pi + \tan^{-1}\left(\frac{4}{-4}\right) \\ &= \pi + \left(-\frac{\pi}{4}\right) \\ &= \pi - \frac{\pi}{4} \\ &= \frac{3\pi}{4} \end{aligned}$$

So, $(r, \theta) = \left(\sqrt{32}, \frac{3\pi}{4}\right)$

Ans:—



~~Incorrect~~

$\sqrt{(10) + (10)}$
 $\sqrt{(10) + (10)}$
 $\sqrt{(10) + (10)}$
 $\sqrt{(10) + (10)}$
 $\sqrt{(10) + (10)}$
 $\sqrt{(10) + (10)}$

03/04/23
MA: 2nd 2

Vectors
non

B → terminal point

$\vec{AB} = \vec{u}$

A
↓
initial point

if $A = (\hat{i} + 2\hat{k})$
 $B = (-\hat{i} + 2\hat{j} + 4\hat{k})$
 $\vec{u} = \vec{AB}$

* if Given the point $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ the Vector $\vec{a}$ with represent $\vec{AB}$ is $\vec{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$ or $(x_2 - x_1)\hat{i}, (y_2 - y_1)\hat{j}, (z_2 - z_1)\hat{k}$.

Laws
non

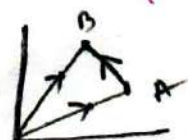
1. Magnitude of $\vec{a} = |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

2. Unit Vectors $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

Q: Find the Vectors represent by the directed line segment with initial position $A(2, -3, 4)$ and terminal position $B(-2, 1, 1)$

⇒ Let $\vec{OA} = 2\hat{i} + (-3\hat{j}) + 4\hat{k}$
 $\vec{OB} = -2\hat{i} + \hat{j} + \hat{k}$

So, $\vec{AB} = \vec{a} = \vec{OB} - \vec{OA}$



$\vec{OA} + \vec{AB} = \vec{OB}$
 $\vec{AB} = \vec{OB} - \vec{OA}$

$$\vec{a} = (-2\hat{i} + \hat{j} + \hat{k}) - (2\hat{i} - 3\hat{j} + 4\hat{k})$$

$$= -4\hat{i} + 4\hat{j} - 3\hat{k}$$

Dot product

Scalar product

if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then Angle between $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Q:- if $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

- What is the magnitude of $\vec{a}$.
- Find the unit vector of $\vec{a}$.
- Find the dot product of $\vec{a}$ and $\vec{b}$.

⇒

is given,

$$\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$$

we know,

magnitude of $\vec{a} = |\vec{a}|$

$$= \sqrt{(2)^2 + (-1)^2 + (3)^2}$$

$$= \sqrt{14}$$

Ans:-

(b)

is given,
~~now~~

$$\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$$

So, Unit vector along $\vec{a}$ $\Rightarrow \hat{a} = \frac{\vec{a}}{|\vec{a}|}$

$$= \frac{2\hat{i} - \hat{j} + 3\hat{k}}{\sqrt{14}}$$

(c)

is given,
~~now~~

$$\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$$

we know:-

$$\begin{aligned} \text{dot product of } a \text{ and } b &\Rightarrow \vec{a} \cdot \vec{b} = (2\hat{i} - \hat{j} + 3\hat{k}) \cdot (\hat{i} - 2\hat{j} + \hat{k}) \\ &= 2 + 2 + 3 \\ &= 7 \end{aligned}$$

P-812:

Q: (17) Find the angle between the two vectors

$$\vec{a} = \langle 1, -4, 1 \rangle \text{ and } \vec{b} = \langle 0, 2, -2 \rangle$$

⇒ We know;

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \quad \text{--- (1)}$$

is given:

$$\vec{a} = \langle 1, -4, 1 \rangle$$

$$\vec{b} = \langle 0, 2, -2 \rangle$$

So, $\vec{a} \cdot \vec{b} = \langle 1, -4, 1 \rangle \cdot \langle 0, 2, -2 \rangle$

$$= 0 - 8 - 2$$

$$= -10$$

$$|\vec{a}| = \sqrt{1^2 + (-4)^2 + 1^2}$$

$$= \sqrt{18}$$

$$|\vec{b}| = \sqrt{0^2 + 2^2 + (-2)^2}$$

$$= \sqrt{8}$$

Now, Substitute the all value in eq (1) ⇒

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)$$

$$= \cos^{-1} \left(\frac{-10}{\sqrt{18} \times \sqrt{8}} \right)$$

$$= 146.5^\circ$$

Ans:

Perpendicular/orthogonal Vectors:—

≡ if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
 $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

the vector are perpendicular/orthogonal
 if $\boxed{\vec{a} \cdot \vec{b} = 0}$

Parallel Vectors:—

≡ if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
 $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

the vector are parallel if

or $\boxed{\vec{a} \times \vec{b} = 0}$

where $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

pg 813
non

23) Determine whether Vectors $\vec{a} = -8\hat{i} + 12\hat{j} + 4\hat{k}$ and $\vec{b} = 6\hat{i} - 9\hat{j} - 3\hat{k}$ are orthogonal, parallel or neither.

$\Rightarrow$ is given,

$$\vec{a} = -8\hat{i} + 12\hat{j} + 4\hat{k}$$

$$\vec{b} = 6\hat{i} - 9\hat{j} - 3\hat{k}$$

we know:-

① If $\vec{a} \cdot \vec{b} = 0$; it's a Orthogonal Vector.

$$\vec{a} \cdot \vec{b} = (-8\hat{i} + 12\hat{j} + 4\hat{k}) \cdot (6\hat{i} - 9\hat{j} - 3\hat{k})$$

$$= -48 + (-108) - 12$$

$$= -168$$

$$\neq 0$$

So, it's not a Orthogonal Vector.

② If $\vec{a} \times \vec{b} = 0$; it's a parallel vector.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -8 & 12 & 4 \\ 6 & -9 & -3 \end{vmatrix}$$

$$= \hat{i}(-36 + 36) - \hat{j}(-24 - 24) + \hat{k}(-72 - 72)$$

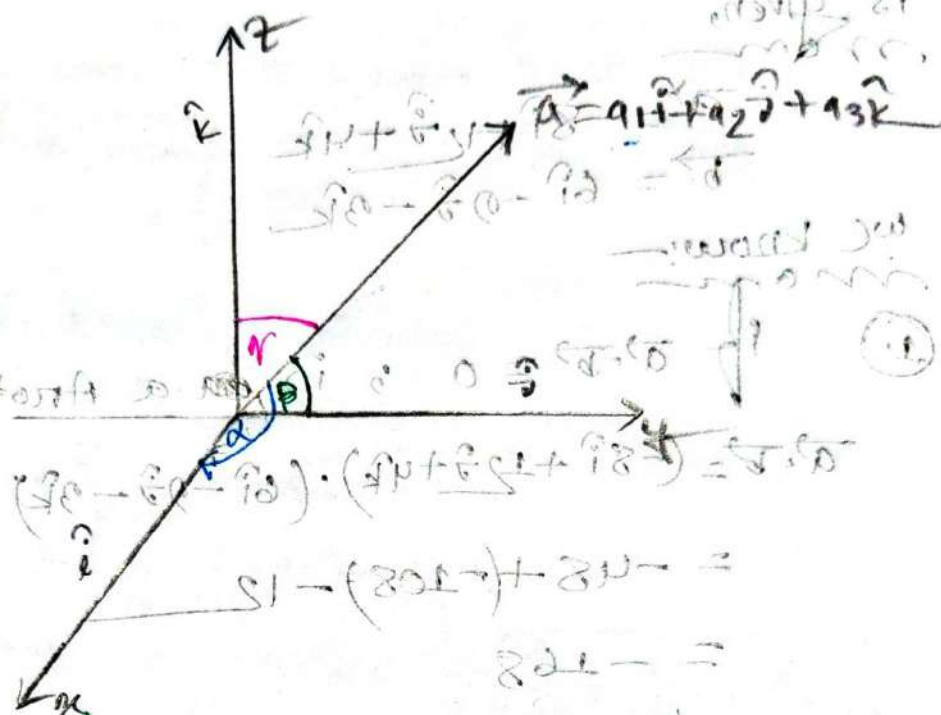
$$= 0\hat{i} - 0\hat{j} + 0\hat{k}$$

$$= 0$$

So, it's a parallel vector.

Ans

Direction Cosine and Direction of Angles



Direction of Cosines

$$1. \cos \alpha = \frac{a_1}{|\vec{A}|}$$

$$2. \cos \beta = \frac{a_2}{|\vec{A}|}$$

$$3. \cos \gamma = \frac{a_3}{|\vec{A}|}$$

Derivation:

$$\vec{A} \cdot \vec{i} = |\vec{A}| |\vec{i}| \cos \alpha$$

$$a_1 \hat{i} = |\vec{A}| \hat{i} \cos \alpha$$

$$a_1 = |\vec{A}| \cos \alpha$$

$$(a_1 - |\vec{A}| \cos \alpha) \hat{i} + (a_2 - |\vec{A}| \cos \beta) \hat{j} + (a_3 - |\vec{A}| \cos \gamma) \hat{k} = \vec{0}$$

$$\vec{0} + \vec{0} + \vec{0} = \vec{0}$$

Direction of Angle:

$$\alpha = \cos^{-1} \left(\frac{a_1}{|\vec{a}|} \right)$$

$$\beta = \cos^{-1} \left(\frac{a_2}{|\vec{a}|} \right)$$

$$\gamma = \cos^{-1} \left(\frac{a_3}{|\vec{a}|} \right)$$

Pr. 8.13:

Q. find the direction cosine and direction angle of the Vector $\vec{r} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

$$\Rightarrow \text{Let, } \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$|\vec{r}| = \sqrt{(1)^2 + (2)^2 + (3)^2}$$

$$= \sqrt{14}$$

Now, we know:

$$\cos \alpha = \frac{a_1}{|\vec{r}|}$$

$$= \frac{1}{\sqrt{14}}$$

$$\alpha = \cos^{-1} \left(\frac{1}{\sqrt{14}} \right)$$

$$= 74.49^\circ$$

$$\cos \beta = \frac{a_2}{|\vec{r}|} = \frac{2}{\sqrt{14}}$$

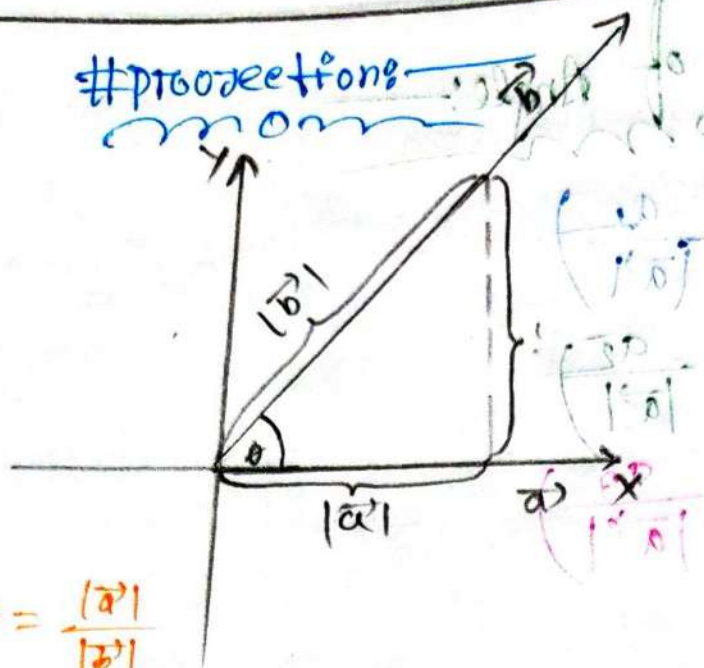
$$\beta = \cos^{-1} \left(\frac{2}{\sqrt{14}} \right)$$

$$= 122.31^\circ$$

$$\cos \gamma = \frac{a_3}{|\vec{r}|} = \frac{3}{\sqrt{14}}$$

$$\gamma = \cos^{-1} \left(\frac{3}{\sqrt{14}} \right)$$

$$= 143.30^\circ$$



$$\cos \theta = \frac{|a|}{|b|}$$

$$|a| = \cos \theta \times |b| \quad \text{--- (1)}$$

Use known: two sides and one angle

$$|a| \cdot |b| \cos \theta = a \cdot b$$

So, Law:

Scalar projection:

1. the scalar projection of $\vec{a}$ onto $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

2. the scalar projection of $\vec{b}$ onto $\vec{a} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

Vector projection:

1. The Vector projection of $\vec{a}$ onto $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \times \frac{\vec{b}}{|\vec{b}|}$

2. The Vector projection of $\vec{b}$ onto $\vec{a} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|} \times \frac{\vec{a}}{|\vec{a}|}$

Q. 14. Find the projection of $\vec{b} = 5\hat{i} - \hat{j}$ onto $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

(14) Find the projection of $\vec{b} = 5\hat{i} - \hat{j}$ onto $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

is given:

$$\vec{b} = 5\hat{i} - \hat{j}$$

$$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{a} \cdot \vec{b} = (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (5\hat{i} - \hat{j})$$

$$= 5 + 0 - 3 = 2$$

we know:

the scalar projection of $\vec{b}$ onto $\vec{a}$ is

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

$$= \frac{2}{\sqrt{1^2 + 2^2 + 3^2}}$$

$$= \frac{2}{\sqrt{14}}$$

the vector projection of $\vec{b}$ onto $\vec{a}$ is

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$

$$= \frac{2}{14} (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= \frac{\hat{i} + 2\hat{j} + 3\hat{k}}{7}$$

Ans.

Q:- Q1 Find the work done by a force $\vec{F} = 8\hat{i} - 6\hat{j} + 9\hat{k}$ that moves an object from the point $(0, 10, 8)$ to point $(6, 12, 20)$ along a straight line. that distance is measured in meter. and the force in Newton.

⇒ Applying the math law

We have:-

$$\vec{F} = 8\hat{i} - 6\hat{j} + 9\hat{k}$$

$$\text{Let } \vec{a} = 0\hat{i} + 10\hat{j} + 8\hat{k}$$

$$\vec{b} = 6\hat{i} + 12\hat{j} + 20\hat{k}$$

$$\vec{ab} = (0+6)\hat{i} + (-10+12)\hat{j} + (-8+20)\hat{k}$$

$$= +6\hat{i} + 2\hat{j} + 12\hat{k}$$

We know:-

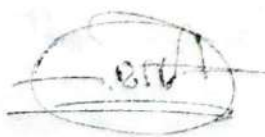
$$W = \vec{F} \cdot \vec{d}$$

$$= \langle 8, -6, 9 \rangle \cdot \langle 6, 2, 12 \rangle$$

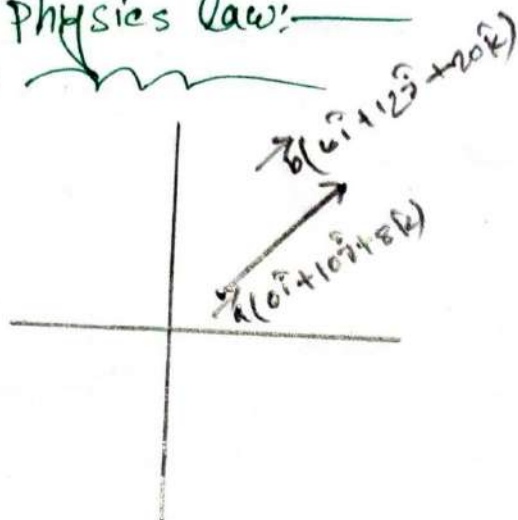
$$= 48 - 12 + 108$$

$$= 144 \text{ J}$$

Ans:-



Q Applying Physics Law:-



we have:-

$$\vec{F} = 8\hat{i} - 6\hat{j} + 0\hat{k}$$

$$\text{let } \vec{a} = (0\hat{i} + 10\hat{j} + 8\hat{k})$$

$$\vec{b} = (6\hat{i} + 12\hat{j} + 20\hat{k})$$

$$\vec{ab} = \vec{b} - \vec{a}$$

$$= (6\hat{i} + 12\hat{j} + 20\hat{k}) - (0\hat{i} + 10\hat{j} + 8\hat{k})$$

$$= 6\hat{i} + 2\hat{j} + 12\hat{k}$$

$$|\vec{F}| = \sqrt{(8)^2 + (-6)^2 + (0)^2} = \sqrt{181}$$

$$|\vec{ab}| = \sqrt{(6)^2 + (2)^2 + (12)^2} = 2\sqrt{46} = D$$

we know:-

$$W = FD \cos \theta$$

$$= \sqrt{181} \times 2\sqrt{46} \times 1$$

$$= 182.40 \text{ J}$$

Ans:-

Cross Product / Vector product

if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then the cross product between $\vec{a}$ and $\vec{b}$ vector is denoted by $\vec{a} \times \vec{b}$ and defined by

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

— expand w.r.t $\hat{i}$

$$\Rightarrow \hat{i}(a_2b_3 - a_3b_2) - \hat{j}(a_1b_3 - a_3b_1) + \hat{k}(a_1b_2 - a_2b_1)$$

$$(\hat{j}a_3 + \hat{i}a_1 + \hat{k}a_2) - (\hat{j}a_3 + \hat{i}a_1 + \hat{k}a_2) =$$

Applications

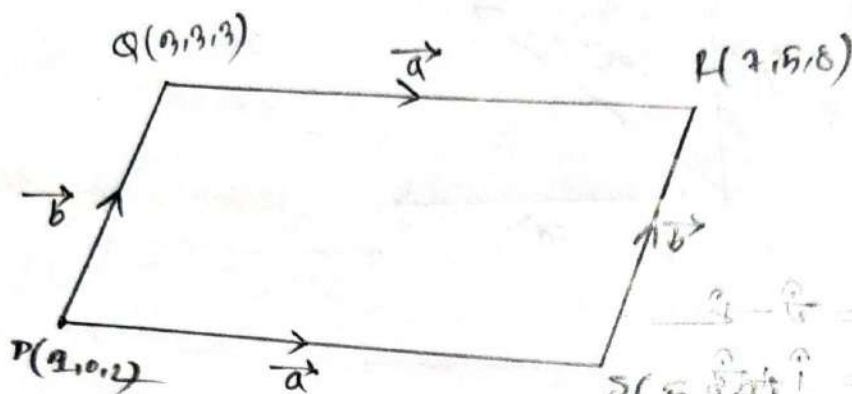
(1)

$\Rightarrow$ The area of parallelogram with sides as two vectors $\vec{a}$ and $\vec{b}$ is the magnitude of cross product $\vec{a} \times \vec{b}$.

$$\text{Area} = |\vec{a} \times \vec{b}|$$

28) Find the area of the parallelogram with vertices
 $P(1, 0, 2)$; $Q(4, 3, 3)$; $R(7, 5, 8)$; $S(5, 2, 7)$

$\Rightarrow$ Let,



~~$\vec{a} = \vec{PS}$~~

$$\vec{a} = \vec{PQ} = 4\hat{i} + 3\hat{j} + 3\hat{k}$$

$$\vec{b} = \vec{PR} = 6\hat{i} + 5\hat{j} + 6\hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 3 \\ 6 & 5 & 6 \end{vmatrix}$$

$$= \hat{i}(2-15) - \hat{j}(4-18) + \hat{k}(12-14)$$

$$= -13\hat{i} + 14\hat{j} - 2\hat{k}$$

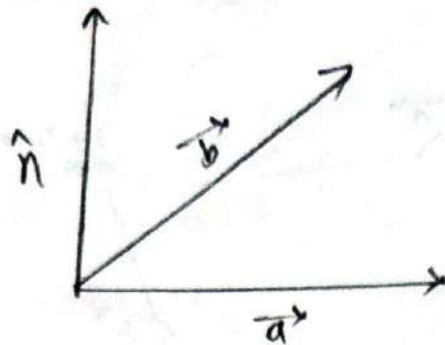
$$|\vec{a} \times \vec{b}| = \sqrt{(-13)^2 + (14)^2 + (-2)^2}$$

$$= 16.40 \text{ unit}^2$$

the area of parallelogram is 16.40 unit^2

[Signature]

(20) Find the unit vector perpendicular to both $\hat{j} - \hat{k}$ and $\hat{i} + \hat{j}$.



$\Rightarrow$ Let, $\vec{a} = \hat{j} - \hat{k}$
 $\vec{b} = \hat{i} + \hat{j}$

$$\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \quad \text{--- (1)}$$

So,

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= \hat{i}(0+1) - \hat{j}(0+1) + \hat{k}(0-1)$$

$$= \hat{i} - \hat{j} - \hat{k}$$

And, $|\vec{a} \times \vec{b}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

Now, substitute the values in equation (1)

$$\hat{n} = \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$$

Scalar Triple Product:

$$\text{if } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$$

Then, Scalar triple product of these vectors.

$$\text{is } \boxed{\vec{a} \cdot (\vec{b} \times \vec{c}) \text{ or } (\vec{a} \times \vec{b}) \cdot \vec{c}}$$

Applications:

Law 1:

1. The Volume of the parallelepiped can be determined by the magnitude of the scalar triple product of the vectors $\vec{a}, \vec{b}, \vec{c}$ that is,

$$\text{Volume } \boxed{V = |\vec{a} \cdot (\vec{b} \times \vec{c})|}$$

$$\text{or } V = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

$$\text{or } V = |(\vec{a} \times \vec{c}) \cdot \vec{b}|$$

2. If the Volume of the parallelepiped determined by the vectors $\vec{a}, \vec{b}, \vec{c}$ is zero, then vectors must lie on the same plane. That is they are coplanar.

Conditions—

$$1. \boxed{\vec{a} \cdot (\vec{b} \times \vec{c}) = 0}$$

$$2. (\vec{a} \times \vec{b}) \cdot \vec{c} = 0$$

$$3. (\vec{a} \times \vec{c}) \cdot \vec{b} = 0$$

Q:- Find the volume of the parallelepiped determined by the vectors $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$.

$$\Rightarrow \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(1-0) - \hat{j}(1-0) + \hat{k}(1-0)$$

$$= \hat{i} - \hat{j} + \hat{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + \hat{k})$$

$$= 1 - 1 + 1$$

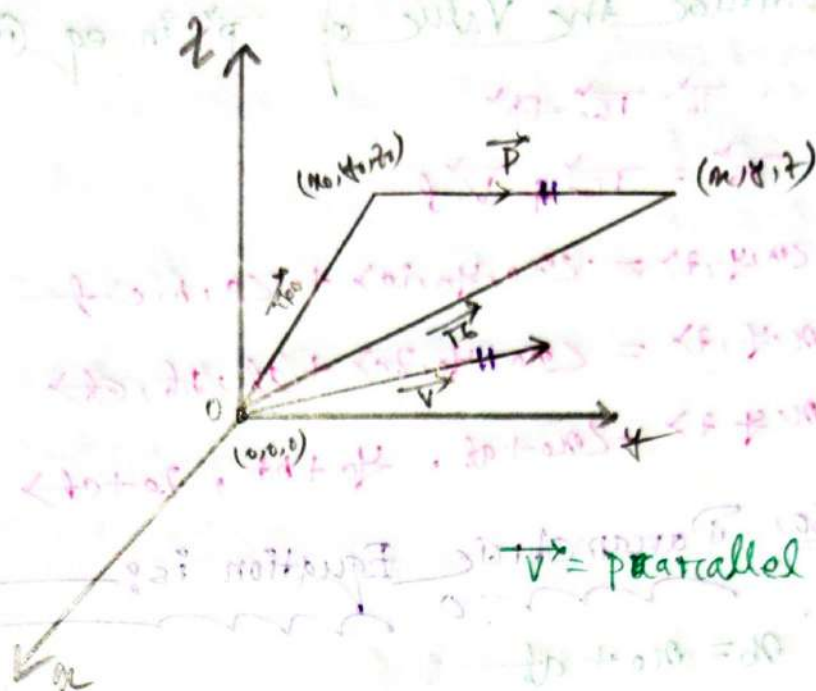
So, Volume $V = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$

$$= 1$$

Ans

Equation of
Lines, in 3D.

Equation of line in 3D



We know, 3-Parallel Positions

if $\vec{a} = \langle a_1, a_2, a_3 \rangle$
 $\vec{b} = \langle b_1, b_2, b_3 \rangle$

then, $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \lambda$

so, $\frac{\vec{r}}{\vec{v}} = \lambda$

$\vec{r} = \vec{v} \lambda$

in the EOL in 3D:

$\vec{r}_{00} + \vec{r} = \vec{r}_0$

where, $\vec{r}_{00} = \langle x_0, y_0, z_0 \rangle$
 $\vec{r} = \langle x, y, z \rangle$

If $\vec{v} = \langle a, b, c \rangle$ then it will be nothing but

Substitute the Value of $\vec{v}$ in eq (1) $\Rightarrow$

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\vec{r} = \vec{r}_0 + \vec{v}t$$

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + \langle at, bt, ct \rangle$$

$$\langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + \langle at, bt, ct \rangle$$

$$\langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$$

So, Parametric Equation is:

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

and Symmetric Equation is:

$$t = \frac{x - x_0}{a}$$

$$t = \frac{y - y_0}{b}$$

$$t = \frac{z - z_0}{c}$$

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Q8. P. 82, 12. (2) Find a Vector equation and parametric equation for the line. The line through the point $(6, -5, 2)$ and parallel to the Vector $\langle 1, 3, -\frac{2}{3} \rangle$

is given,

$$\vec{r}_0 = \langle 6, -5, 2 \rangle$$

$$\vec{r} = \langle x, y, z \rangle$$

$$\vec{v} = \langle 1, 3, -\frac{2}{3} \rangle$$

We know

The Vector equation is:

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\langle x, y, z \rangle = \langle 6, -5, 2 \rangle + t \langle 1, 3, -\frac{2}{3} \rangle$$

$$\langle x, y, z \rangle = \langle 6+t, -5+3t, 2-\frac{2t}{3} \rangle$$

The Parametric equation is:

$$x = 6+t$$

$$y = -5+3t$$

$$z = 2-\frac{2t}{3}$$

The Symmetric equation is:

$$t = \frac{x-6}{1}$$

$$t = \frac{z-2}{-\frac{2}{3}}$$

$$\frac{x-6}{1} = \frac{z-2}{-\frac{2}{3}} \Rightarrow \frac{x-6}{1} = \frac{3(2-z)}{2} \Rightarrow \frac{x-6}{1} = \frac{6-3z}{2}$$

Q: Find the parametric equation and Symmetric equation for the line through the point $(-8, 1, 4)$ and $(3, -2, 4)$

⇒ So, Parallel Vector $\vec{V} = \langle 3+8, -2-1, 4-4 \rangle$

$$\vec{V} = \langle 11, -3, 0 \rangle$$

$$\vec{r}_0 = \langle -8, 1, 4 \rangle$$

$$\vec{r} = \vec{r}_0 + t\vec{V}$$

$$\vec{r} = \vec{r}_0 + t\vec{V}$$

$$\langle x, y, z \rangle = \langle -8, 1, 4 \rangle + t\langle 11, -3, 0 \rangle$$

$$\langle x, y, z \rangle = \langle -8 + 11t, 1 - 3t, 4 + 0t \rangle$$

So, the parametric equation is:

$$x = -8 + 11t$$

$$y = 1 - 3t$$

$$z = 4$$

and, the symmetric equation is:

$$t = \frac{x+8}{11}$$

$$t = \frac{y-1}{-3}$$

$$\frac{x+8}{11} = \frac{y-1}{-3}, \quad z=4$$

Q. 10 Find the Parametric equation and symmetric equation through the point $(2, 1, 0)$ and perpendicular to both $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$

⇒ Let,

$$\vec{a} = \hat{i} + \hat{j}$$

$$\vec{b} = \hat{j} + \hat{k}$$

$$\text{So, } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(1-0) - \hat{j}(1-0) + \hat{k}(1-0)$$

$$\vec{V} = \hat{i} - \hat{j} + \hat{k}$$

is given —

$$\vec{r_0} = \langle 2, 1, 0 \rangle$$

we know —

The Vector equation is —

$$\vec{r} = \vec{r_0} + V \cdot t$$

$$\langle x, y, z \rangle = \langle 2, 1, 0 \rangle + \langle 1, -1, 1 \rangle t$$

$$\langle x, y, z \rangle = \langle 2+t, 1-t, 0+t \rangle$$

P.T.O

The Parametric equation is

$$x = 2 + t$$

$$y = 1 - t$$

$$z = t$$

The Symmetric equation is

$$t = x - 2$$

$$t = 1 - y$$

$$t = z$$

$$\frac{x-2}{1} = \frac{1-y}{1} = \frac{z}{1}$$

Check 2 lines are parallel

Skew, intersecting

$$\text{if } L_1 = \frac{x-e}{a_1} = \frac{y-e}{b_1} = \frac{z-e}{c_1}$$

$$L_2 = \frac{x-e}{a_2} = \frac{y-e}{b_2} = \frac{z-e}{c_2}$$

$$\text{so, } \vec{v}_1 = \langle a_1, b_1, c_1 \rangle$$

$$\vec{v}_2 = \langle a_2, b_2, c_2 \rangle$$

so, the condition of parallel, skew,

intersecting
D.T.O

is: —

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

if the ratios are equal, the lines are parallel. (1)

Q1) Determine whether the lines L_1 and L_2 are parallel, skew, or intersecting. If they intersect find the point of intersection.

$$L_1 = \frac{x-2}{1} = \frac{y-3}{2} = \frac{z-1}{-3}$$

$$L_2 = \frac{x-3}{1} = \frac{y+4}{3} = \frac{z-2}{-7}$$

$$\Rightarrow \text{So, } \vec{v}_1 = \langle 1, 2, -3 \rangle$$

$$\vec{v}_2 = \langle 1, 3, -7 \rangle$$

If we know:
no

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

then two lines are parallel.

Now $\frac{1}{1} \neq \frac{2}{3} \neq \frac{-3}{-7}$

So, not parallel.

Ans:

Exercise for practice:—

① Find the Vector equation and parametric equations, for the line which passes through the point $(0, 14, -10)$ and parallel to the line

$$x = -1 + 2t, y = 6 - 3t, z = 3 + 2t$$

⇒ is given:—

$$\vec{r}_0 = (0\hat{i} + 14\hat{j} - 10\hat{k}) = \langle 0, 14, -10 \rangle$$

and $\vec{v}$ from the line is:—

$$\vec{v} = \langle 2, -3, 2 \rangle$$

We know:—

The Vector equation:—

$$\vec{r} = \vec{r}_0 + \vec{v}t$$

$$\langle x, y, z \rangle = \langle 0, 14, -10 \rangle + \langle 2, -3, 2 \rangle t$$

$$\langle x, y, z \rangle = \langle 0 + 2t, 14 - 3t, -10 + 2t \rangle$$

The parametric equation is:—

$$x = 2t$$

$$y = 14 - 3t$$

$$z = -10 + 2t$$

(2) Find the parametric equation and symmetric equations for the line through the origin and the point $(4, 3, -1)$

⇒ is given —

$$\vec{r}_0 = \langle 0, 0, 0 \rangle$$

$$\text{So, } \vec{v} = \langle 4-0, 3-0, -1-0 \rangle = \langle 4, 3, -1 \rangle$$

$$\vec{v} = \langle 4, 3, -1 \rangle$$

$$\vec{r} = \langle x, y, z \rangle$$

We know: —

(1) The Vector equation is:

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\langle x, y, z \rangle = \langle 0, 0, 0 \rangle + t\langle 4, 3, -1 \rangle$$

$$\langle x, y, z \rangle = \langle 0+4t, 0+3t, 0-t \rangle$$

(2) The Parametric equation is:

$$x = 4t$$

$$y = 3t$$

$$z = -t$$

(3) The Symmetric equation is:

$$t = \frac{x}{4}, t = \frac{y}{3}, t = -z$$

$$\frac{x}{4} = \frac{y}{3} = \frac{-z}{1}$$

Ans.

③ Find the Vector equation and parametric equation for the line which passes through the point $(-6, 2, 3)$ and parallel to the line $\frac{x}{2} = \frac{y}{3} = z+1$

$\Rightarrow$ is given—

$$\vec{r}_0 = \langle -6, 2, 3 \rangle$$

$$\text{So, } \vec{v} = \langle 2, 3, 1 \rangle$$

$$\vec{r} = \langle x, y, z \rangle$$

We know—

The Vector equation is—

$$\vec{r} = \vec{r}_0 + \vec{v}t$$

$$\langle x, y, z \rangle = \langle -6, 2, 3 \rangle + \langle 2, 3, 1 \rangle t$$

$$\langle x, y, z \rangle = \langle -6 + 2t, 2 + 3t, 3 + t \rangle$$

The parametric equation is—

$$x = -6 + 2t$$

$$y = 2 + 3t$$

$$z = 3 + t$$

Ans:-

④ Determine whether the lines L_1 and L_2 are parallel, skew or intersecting. if they intersect find the point of intersection.

$$L_1 = \underline{x = 3 + 2t}, \underline{y = 4 - t}, \underline{z = 1 + 3t}$$

$$L_2 = \underline{x = 1 + 4s}, \underline{y = 3 - 2s}, \underline{z = 4 + 5s}$$

⇒ is given:—

$$\underline{\vec{v}_1 = \langle 2, -1, 3 \rangle}$$

$$\underline{\vec{v}_2 = \langle 4, -2, 5 \rangle}$$

We know:—

The lines L_1 and L_2 are parallel; skew, or intersecting if $\underline{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$

So, that,

$$\underline{\frac{2}{4} = \frac{-1}{-2} = \frac{3}{5}}$$

$$\Rightarrow \underline{\frac{1}{2} = \frac{1}{2} = \frac{3}{5}}$$

Since these ratios are not equal so the L_1 and L_2 line are not parallel.

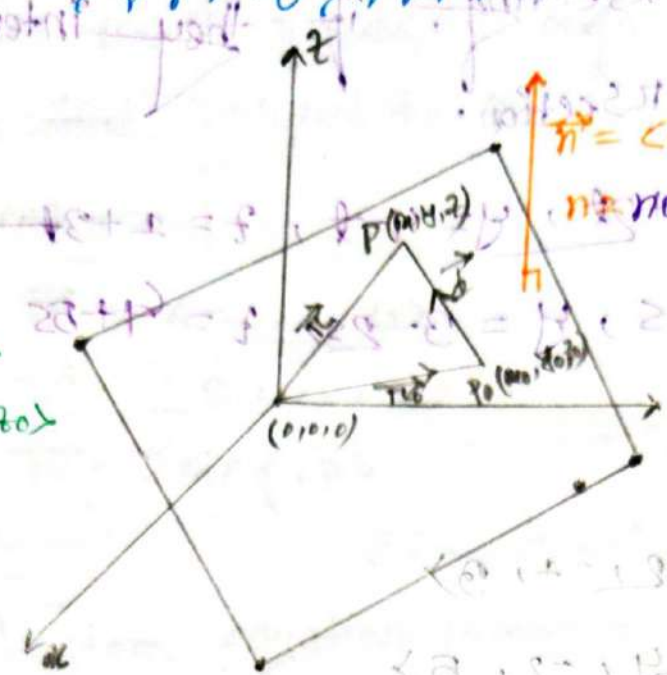
Ans:—

Equation of plane

here:-

$$\vec{r} = \langle x, y, z \rangle$$

$$\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$$



$\vec{n} = \langle a, b, c \rangle$
normal Vector / Perpendicular Vector

$$\vec{r}_0 + \vec{d} = \vec{r}$$

$$\vec{d} = \vec{r} - \vec{r}_0$$

Since, $\vec{n}$ are perpendicular of $\vec{d}$,

we know:-

$$\vec{n} \cdot \vec{d} = 0$$

Substitute the Value of $\vec{n} \cdot \vec{d}$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

This is the equation of plane.

PROBLEM Find the equation of plane through the point $(5, 3, 5)$ with Normal vector $2\hat{i} + \hat{j} - \hat{k}$.

$\Rightarrow$ is given,

$$\vec{r}_0 = \langle 5, 3, 5 \rangle$$

$$\vec{n} = \langle 2, 1, -1 \rangle$$

We know

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$2(x - 5) + 1(y - 3) - 1(z - 5) = 0$$

$$2x - 10 + y - 3 - z + 5 = 0$$

$$\boxed{2x + y - z = 8}$$

Ans:-

26 Find the equation of the plane through the point $(2, 0, 1)$ and perpendicular to the line $x = 3t$, $y = 2 - t$, $z = 3 + 4t$.

$\Rightarrow$ is given,

$$\vec{r}_0 = \langle 2, 0, 1 \rangle$$

$$\vec{n} = \langle 3, -1, 4 \rangle$$

We know

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$3(x - 2) - 1(y - 0) + 4(z - 1) = 0$$

$$3x - 6 - y + 4z - 4 = 0$$

$$\boxed{3x - y + 4z = 10}$$

Ans:-

Q2) Find the equation of the plane through the point $(1, -1, -1)$ and parallel to the plane $5x - y - z = 6$

is given—

$$\vec{r_0} = \langle 1, -1, -1 \rangle$$

$$\vec{n} = \langle 5, -1, -1 \rangle$$

We know—

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$5(x - 1) + (-1)(y + 1) + (-1)(z + 1) = 0$$

$$5x - 5 - y - 1 - z - 1 = 0$$

$$5x - y - z = 7$$

Ans:-

Q3) Find the equation of the plane through the point $(3, 0, -1)$, $(-2, -2, 3)$ and $(7, 1, -4)$

Let, $\vec{p} = \langle 3, 0, -1 \rangle$

$$\vec{q} = \langle -2, -2, 3 \rangle$$

$$\vec{r} = \langle 7, 1, -4 \rangle$$

$$\vec{a} = \vec{r} - \vec{p} = \langle 7, 1, -4 \rangle - \langle 3, 0, -1 \rangle$$

$$= \langle 4, 1, -3 \rangle$$

$$\vec{b} = \vec{q} - \vec{p} = \langle -2, -2, 3 \rangle - \langle 3, 0, -1 \rangle$$

$$= \langle -5, -2, 4 \rangle$$



Since both $\vec{a}$ and $\vec{b}$ lie in the same plane their cross product $\vec{a} \times \vec{b}$ is orthogonal to the plane and can be taken as the Normal Vector.

$$\begin{aligned}\vec{n} &= \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & -3 \\ -5 & -2 & 4 \end{vmatrix} \\ &= \hat{i}(4-15) - \hat{j}(16-15) + \hat{k}(-8+5) \\ &= -11\hat{i} - \hat{j} - 3\hat{k}\end{aligned}$$

Now: $\vec{r_0} = (3, 0, -1)$

$\vec{n} = (-11, -1, -3)$

We know:—

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$-11x + 6 - y - 3z - 3 = 0$$

$$-11x - y - 3z = -9$$

$$11x + y + 3z = 9$$

Q3) Determine whether the planes are parallel, perpendicular or neither. if neither then find the angle between them.

$$x + 2y - z = 2 \quad \text{and} \quad 2x - 2y + z = 1$$

⇒ Check

Parallel

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

$$\frac{1}{2} = \frac{2}{-2} = \frac{-1}{1}$$

$$\frac{1}{2} \neq -1 = -1$$

not

Parallel

check

~~Perpendicular.~~

$$\vec{n}_1 \cdot \vec{n}_2 = \langle 1, 2, -1 \rangle \cdot \langle 2, -2, 1 \rangle$$

$$= 2 - 4 - 1 = -3 \neq 0$$

Not

~~Perpendicular.~~

So gts

neither.

Now:-

we know:-

$$\theta = \cos^{-1} \left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} \right)$$

$$= \cos^{-1} \left(\frac{-3}{\sqrt{1^2+2^2+1^2} \sqrt{2^2+2^2+1^2}} \right)$$

$$= 114.09^\circ$$

Ans:-

the angle between the two planes is 114.09°

Ans:-